TORMS — Complete System Specification

Field	Value
Product	TORMS — Transcript & Official Records Management System
Document type	Complete System Specification (SRS + Architecture + Features + LLD)
Date	November 2026
Classification	Proprietary

Document contents

1. System Requirements Specification (SRS)
2. Architecture Design
3. Features & Requirements
4. Low-Level Design (LLD) Workflows

Part 1 — System Requirements Specification (SRS)

Transcript & Official Records Management System (TORMS)

Field	Value
Document ID	SRS-TORMS-001
Product	TORMS (Transcript & Official Records Management System)
Repository	`WMSU/wmsu-platform` (`apps/records-web` , `apps/base-api` module `records`)
Version	1.0
Date	November 2026
Classification	Proprietary

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1. Introduction

1.1 Purpose

This Software Requirements Specification defines functional and non-functional requirements for **TORMS**, the official academic records and certification system managing transcript of records (TOR) issuance, certification document processing, academic ledger management, alumni registry, public document verification, and records office operations for Philippine university compliance.

1.2 Scope

In scope

- TOR request management: creation, requirement tracking, payment verification linkage, status workflow, document generation, issuance logging, print audit trail
- Certification types and certification request processing: honorable dismissal, good moral character, units earned, completion certificates, diploma verification
- Public document verification: unauthenticated QR code and reference number verification with tamper-evident audit trails
- Academic ledger projections synchronized from USiS enrollment system (grades, enrollment history)
- Ledger discrepancy detection and reconciliation workflows
- Alumni registry: profile management, standing classification, graduation-driven enrollment
- Records officer dashboard and comprehensive reporting (issuance statistics, request aging, alumni demographics)
- University Platform Access SSO integration (platform code `records`)
- Student self-service request submission and status tracking

- Document numbering schemes and template management
- Integration with UFMS for payment/clearance verification where configured

Out of scope

- Authoritative grade entry and enrollment workflow (owned by **USiS**)
- Tuition assessment, cashiering, official receipt issuance (owned by **UFMS**)
- Library operations, document repository (owned by **ULMIS**, **UDMS**)
- Inventory and asset management (owned by **UIMS**)
- Student admissions and registration approval (owned by **USiS**)

1.3 Definitions

Term	Definition
TORMS	Transcript & Official Records Management System
TOR	Transcript of Records — official academic transcript document
Issuance	Generated official document with verification metadata, print log, and QR code
Ledger projection	TORMS copy of academic grades and enrollment history synchronized from USiS (USiS remains system of record)
Certification	Official non-TOR document types: honorable dismissal, good moral character, units earned, completion, diploma verification
Verification token	Unique code (UUID or sequential) enabling public unauthenticated verification of issued documents
Discrepancy	Mismatch between TORMS ledger projection and USiS source data requiring registrar review
Alumni standing	Classification of alumni status: active, inactive, distinguished, emeritus

1.4 References

- `WMSU/wmsu-platform/README.md`
- `apps/web-portal/public/docs/TORMS-Complete-Guide.html`
- `apps/web-portal/public/docs/TORMS-API-Specs-Manual.html`
- USiS Grade Projection API Specification
- UFMS Internal Clearance API Specification
- University Platform Access Architecture

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2. Overall Description

2.1 Product Perspective

TORMS operates within the WMSU Campus Information System modular architecture:

Component	Technology	Role
Records web	React 19 + Vite	Staff and student UI for requests, issuance, alumni management
Base API records module	AdonisJS 7 + Lucid ORM	REST services at `/api/v1/records/*`
Database	PostgreSQL schema `records`	Official records system of record for issuances and alumni
Identity provider	University Platform Access	Authentication, authorization, user management
Queue system	BullMQ + Redis	Background jobs (sync, notifications, report generation)

Surface	Dev port	Production example
Records web	`http://127.0.0.1:5181`	`https://records.wmsu.edu.ph`
Base API	`http://127.0.0.1:3500`	`https://api.wmsu.edu.ph`
Identity API	`http://127.0.0.1:3411`	`https://auth.wmsu.edu.ph`
Public verify	Records web `/verify`	`https://records.wmsu.edu.ph/verify`

Health endpoint: `GET /health` → `{ status: 'ok', service: 'base-api', module: 'records' }`

2.2 User Classes

Role	Primary capabilities	Access level
Records Officer	Process TOR/certification requests, sync academic data, issue documents, manage alumni, generate reports	Full records management
Student	Submit and track own TOR/certification requests, download issued documents	Own requests only
Registrar (cross-system)	Review sync discrepancies, approve special requests	Records oversight
Alumni	Update contact information, view issued documents	Own alumni profile
Identity Administrator	Full TORMS access when `records` platform granted	All TORMS permissions
Public Visitor	Verify authenticity of issued documents without authentication	Verification only

Frontend permission checks provide user experience; API enforces authorization on every route.

2.3 Constraints

- Platform tile visibility requires Identity platform grant `records`
- Grades are NOT authored in TORMS — all academic data synchronized read-only from USiS
- Sync discrepancies must surface to records officers for manual review and resolution
- Public verification is unauthenticated and rate-limited to prevent abuse
- Internal alumni-from-graduation route requires service API key protection
- TOR and certification numbering must follow configured foundation settings (year-based, sequential)
- Document templates and layout configured per institution requirements
- Historical data integrity preserved — issued documents immutable after generation

2.4 Assumptions

- USiS enrollment system is running and grade data available for synchronization
- Payment/clearance verification checks integrate with UFMS when configured (optional gating)
- Graduation events from USiS enrollment trigger internal alumni profile creation
- Document issuance templates provided in PDF generation format (e.g., Puppeteer/headless Chrome)
- University seal and authorized signatory images configured in system settings
- Records office staff trained on request processing workflows
- Demo/staging environments provision sample students, requests, and issued documents
- Regular automated sync jobs scheduled (nightly grade sync, weekly enrollment history)

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3. Functional Requirements

3.1 Authentication & Authorization

ID	Requirement	Priority
FR-AUTH-001	System shall authenticate all users via University Platform Access SSO	Must
FR-AUTH-002	System shall resolve user home path based on role (records officer → `/staff`, student → `/student`, alumni → `/alumni`)	Must
FR-AUTH-003	System shall enforce `torms.*` permission checks on every authenticated API endpoint	Must
FR-AUTH-004	Public verification routes shall be unauthenticated with rate limiting	Must
FR-AUTH-005	Internal service routes shall require `X-Service-Api-Key` header validation	Must

3.2 TOR Request Management

ID	Requirement	Priority
FR-TOR-001	Students shall submit TOR requests with purpose, delivery mode (pickup/mail), and copies quantity	Must
FR-TOR-002	System shall assign unique request numbers using configured format (e.g., `TOR-2026-0001`)	Must
FR-TOR-003	System shall track request status: submitted → requirements_pending → requirements_complete → processing → approved → issued → released	Must
FR-TOR-004	Records officers shall view all TOR requests with filtering by status, date, student	Must
FR-TOR-005	Records officers shall transition request status with required notes for audit	Must
FR-TOR-006	System shall validate clearance requirements before allowing approval (configurable)	Must
FR-TOR-007	System shall verify payment status from UFMS if payment gating enabled	Should
FR-TOR-008	System shall attach requirement documents (clearance certificate, ID, etc.)	Must
FR-TOR-009	System shall notify students on status transitions via email/SMS when configured	Should
FR-TOR-010	System shall generate TOR PDF document with academic ledger, seal, and signatories	Must
FR-TOR-011	System shall embed verification QR code on generated TOR documents	Must
FR-TOR-012	System shall log all TOR print events with timestamp, officer, and copy count	Must
FR-TOR-013	Students shall download digital copy of issued TOR via portal	Must

3.3 Certification Request Management

ID	Requirement	Priority
FR-CRT-001	System shall support certification types: honorable dismissal, good moral character, units earned, completion, diploma verification	Must
FR-CRT-002	Students shall submit certification requests with type selection and purpose	Must
FR-CRT-003	System shall assign unique certification numbers per type (e.g., `HD-2026-0001`, `GMC-2026-0001`)	Must

FR-CRT-004	System shall track certification status workflow parallel to TOR (submitted → processing → issued → released)	Must
FR-CRT-005	Records officers shall configure certification types with templates and requirements	Should
FR-CRT-006	System shall generate certification PDFs from configured templates with verification codes	Must
FR-CRT-007	System shall log certification print events	Must

3.4 Academic Ledger Sync

ID	Requirement	Priority
FR-LED-001	System shall synchronize grade data from USiS via internal API pull	Must
FR-LED-002	System shall create/update ledger projection records without mutating USiS source	Must
FR-LED-003	System shall detect discrepancies between ledger projection and USiS source	Must
FR-LED-004	System shall surface discrepancy reports to records officers with comparison view	Must
FR-LED-005	Records officers shall manually resolve discrepancies with audit logging	Must
FR-LED-006	System shall synchronize enrollment history (terms, status, programs) from USiS	Must
FR-LED-007	System shall track sync run metadata (timestamp, record count, errors)	Must
FR-LED-008	System shall retry failed sync operations with exponential backoff	Should
FR-LED-009	Records officers shall manually trigger on-demand sync for specific students	Should

3.5 Alumni Registry

ID	Requirement	Priority
FR-ALM-001	System shall create alumni profile automatically from USiS graduation event via internal API	Must
FR-ALM-002	Alumni shall update contact information (address, phone, email) via portal	Must
FR-ALM-003	Alumni shall view issued TOR and certification documents	Must
FR-ALM-004	Records officers shall maintain alumni standing classification (active, inactive, distinguished, emeritus)	Should
FR-ALM-005	System shall track alumni employment and further education (self-reported)	Should

FR-ALM-006	Records officers shall generate alumni demographic reports	Should
FR-ALM-007	System shall support bulk alumni data import from legacy systems	Could

3.6 Public Document Verification

ID	Requirement	Priority
FR-VRF-001	System shall provide public `/verify` page without authentication	Must
FR-VRF-002	Public users shall verify documents by entering verification code or scanning QR	Must
FR-VRF-003	System shall return verification result: valid (with student name, document type, issue date) or invalid	Must
FR-VRF-004	System shall NOT expose sensitive PII beyond name, document type, and issue date	Must
FR-VRF-005	System shall log all verification attempts with IP address and timestamp	Must
FR-VRF-006	System shall rate-limit verification endpoint to prevent scraping (e.g., 10 requests/minute/IP)	Must
FR-VRF-007	System shall display verification result with institutional branding	Should

3.7 Dashboard & Reporting

ID	Requirement	Priority
FR-DSH-001	Records officers shall view dashboard with pending requests, recent issuances, sync status	Must
FR-DSH-002	Dashboard shall display KPIs: total requests by status, average processing time, issuance volume by month	Should
FR-RPT-001	System shall generate TOR issuance report by date range, program, student type	Must
FR-RPT-002	System shall generate certification issuance report by type and date range	Must
FR-RPT-003	System shall generate request aging report (days in each status)	Must
FR-RPT-004	System shall generate alumni demographics report (graduation year, program, location)	Should

FR-RPT-005	System shall export all reports in CSV and PDF formats	Must
FR-RPT-006	System shall log all report exports with user and timestamp for audit	Must

3.8 Configuration Management

ID	Requirement	Priority
FR-CFG-001	Administrators shall configure document numbering schemes (prefix, year format, sequence)	Must
FR-CFG-002	Administrators shall upload document templates (TOR, certifications) in PDF-compatible format	Should
FR-CFG-003	Administrators shall configure authorized signatories with titles and signature images	Must
FR-CFG-004	Administrators shall configure clearance requirement rules per document type	Should
FR-CFG-005	System shall maintain audit log of configuration changes	Must

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4. External Interface Requirements

4.1 Integration Interfaces

Interface	Direction	Protocol	Purpose
University Platform Access	Inbound	OIDC/JWKS	SSO authentication, JWT verification, role resolution
USiS Enrollment API	Inbound	REST+JSON	Grade projection pull, enrollment history sync
USiS Graduation Events	Inbound	REST/Webhook	Alumni profile creation trigger
UFMS Finance API	Outbound (optional)	REST+JSON	Payment/clearance verification for request gating
SMTP/SMS Gateway	Outbound	SMTP/HTTP	Request status notifications
Object Storage	Outbound	S3-compatible	Issued document archival (PDF storage)

4.2 API Specifications

Records API base: `/api/v1/records/*`

Internal service routes: `/api/v1/records/internal/*` (service key protected)

Public routes: `/api/v1/records/public/verify/*` (unauthenticated, rate-limited)

OpenAPI documentation: Available at `/api/v1/records/docs` (Swagger UI)

Authentication: Bearer JWT from University Platform Access in `Authorization` header

4.3 User Interface Requirements

- Responsive design supporting desktop (≥1280px), tablet (768-1023px), and mobile (360-767px)
- Browser compatibility: Chrome/Edge (latest 2), Firefox (latest 2), Safari (latest 2)
- Accessibility: WCAG 2.1 Level AA for public verification page
- Print-optimized PDF generation for TOR and certifications
- QR code rendering in standard formats (PNG, SVG)
- Document preview before issuance
- Bulk request processing interface for records officers

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5. Non-Functional Requirements

5.1 Performance

ID	Requirement	Target
NFR-PERF-001	API response time (p95) for standard queries	≤ 2 seconds
NFR-PERF-002	TOR/certification PDF generation time	≤ 10 seconds
NFR-PERF-003	Public verification response time (p95)	≤ 1 second
NFR-PERF-004	Health check endpoint response time	≤ 500 ms
NFR-PERF-005	Academic ledger sync job completion for 1000 students	≤ 30 minutes

5.2 Scalability

ID	Requirement
NFR-SCAL-001	Support ≥10,000 alumni records
NFR-SCAL-002	Support ≥5,000 TOR requests per academic year
NFR-SCAL-003	Horizontal scaling of API replicas
NFR-SCAL-004	Background job queue processing with worker pools

5.3 Security

ID	Requirement
NFR-SEC-001	TLS 1.2+ required for all client-server communication
NFR-SEC-002	RBAC enforcement on every authenticated API endpoint using `torms.*` permissions
NFR-SEC-003	Service-to-service communication via API keys for internal routes
NFR-SEC-004	Public verification rate limiting (10 requests/minute/IP)
NFR-SEC-005	Document verification tokens cryptographically unique (UUID v4)
NFR-SEC-006	Issued document PDFs stored with restricted access (records officers only)
NFR-SEC-007	Audit logging of all issuance, print, and configuration changes
NFR-SEC-008	CORS allowlist limited to authorized web client origins
NFR-SEC-009	Non-root container execution in production
NFR-SEC-010	Student PII protection per RA 10173 (Data Privacy Act)

5.4 Reliability

ID	Requirement
NFR-REL-001	System availability ≥99% during business hours
NFR-REL-002	Automated database backups (daily incremental, weekly full)
NFR-REL-003	Point-in-time recovery capability ≤15 minutes RPO
NFR-REL-004	Graceful degradation when USiS/UFMS temporarily unavailable
NFR-REL-005	Issued documents immutable — no deletion, only void with audit trail

5.5 Maintainability

ID	Requirement
NFR-MAIN-001	Modular domain architecture with separation of concerns
NFR-MAIN-002	Comprehensive API documentation (OpenAPI 3.0+)
NFR-MAIN-003	Structured logging with correlation IDs
NFR-MAIN-004	Database schema migrations versioned and reversible
NFR-MAIN-005	Document template hot-reloading without deployment

5.6 Compliance

ID	Requirement

NFR-COMP-001	RA 10173 (Data Privacy Act) compliance for student/alumni PII
NFR-COMP-002	FERPA-equivalent protections for educational records
NFR-COMP-003	Audit trail retention ≥7 years per Philippine records management guidelines
NFR-COMP-004	Document issuance chain of custody preservation

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6. Verification Criteria

6.1 Functional Verification

Scenario	Expected result	Test type
Records officer login via SSO	Redirect to `/staff` dashboard	Integration
Student login via SSO	Redirect to `/student` requests page	Integration
Student submit TOR request	Request created with status `submitted`	E2E
Records officer transition request to processing	Status updated, audit log entry created	Unit
Generate TOR PDF	PDF file generated with QR code and ledger data	Integration
Public verify valid TOR code	Returns valid with student name, document type, issue date	API test
Public verify invalid code	Returns invalid/not found	API test
Sync grades from USiS	Ledger projections created/updated, no USiS mutation	Integration
Detect grade discrepancy	Discrepancy record created, visible in officer view	Unit
Alumni graduation trigger	Alumni profile created via internal API	Integration
Rate limit verification endpoint	HTTP 429 after 10 requests in 1 minute	Security test

6.2 Non-Functional Verification

Scenario	Expected result	Test type
100 concurrent public verifications	p95 response ≤1s, no errors	Load test
TOR PDF generation for 50-page transcript	Completes in ≤10s	Performance test
Unauthorized access to internal sync API	HTTP 401/403	Security test
Database failover	Recovery within 15 minutes	DR drill

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7. Document Control

Version	Date	Description
1.0	November 2026	Complete System Specification from wmsu-platform records module

Related Design Documents

- **TORMS-Architecture.md** — Architecture design (containers, domains, security, deployment)
- **TORMS-Features-Requirements.md** — Feature catalog + requirements traceability
- **TORMS-LLD.md** — Low-level design workflows (TOR, certifications, sync, verification, alumni)

End of SRS — TORMS

Part 2 — Architecture Design

TORMS — Transcript & Official Records Management System Architecture

Field	Value
Document ID	AD-TORMS-001
Product	TORMS
Companions	TORMS-SRS.md · TORMS-Features-Requirements.md · TORMS-LLD.md
Date	November 2026

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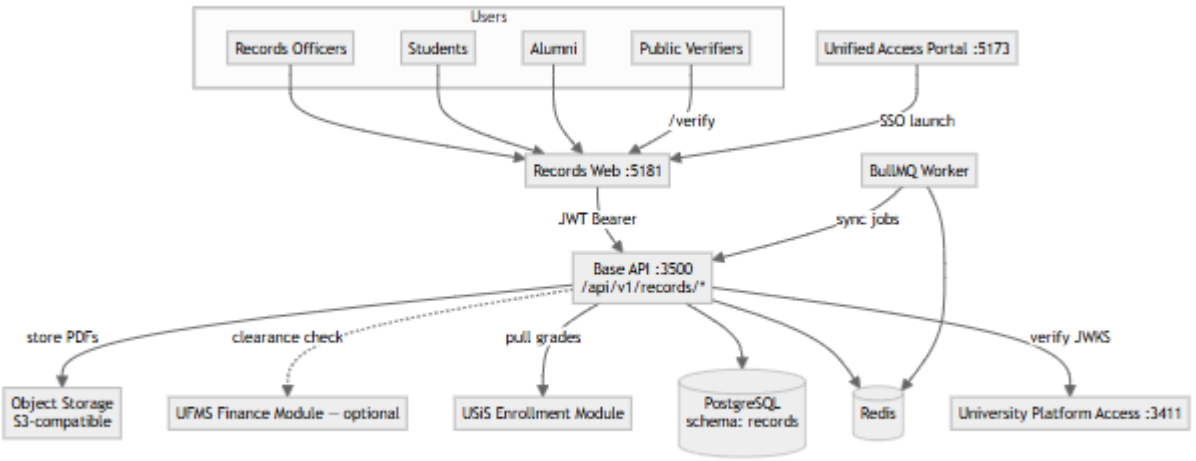
1. Purpose

Architecture design for TORMS within WMSU CIS modular monolith — covering system context, container architecture, domain module organization, security architecture, deployment topology, integration patterns, and quality attributes for official records management.

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2. System Context

TORMS integrates with Identity for SSO, USiS for academic data sync, and optional UFMS for clearance checks:



Key relationships:

- Records officers, students, and alumni authenticate via Portal SSO

- Public verifiers access `/verify` without authentication
- All authenticated requests flow through Base API records module
- USiS is authoritative source for grades; TORMS stores projections
- UFMS provides optional clearance/payment verification
- Background sync jobs run via BullMQ worker
- Issued document PDFs archived to object storage

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3. Container View

Container	Technology	Port	Purpose
Records web	React 19 + Vite	5181	Staff operations, student requests, alumni portal, public verify
Base API (records module)	AdonisJS 7 + Lucid ORM	3500	Domain services, controllers, business logic
PostgreSQL	PostgreSQL 15+	5432	Schema `records` — official records SoR (issuances, alumni)
Redis	Redis 7+	6379	Cache, queue backend, rate limiting
BullMQ Worker	Node.js + BullMQ	N/A	Background sync, notifications, PDF generation
Object Storage	S3-compatible	N/A	Issued document PDF archival
University Platform Access	Identity service	3411	Authentication, authorization
USiS Enrollment	CIS module	3500	Grade/enrollment history source

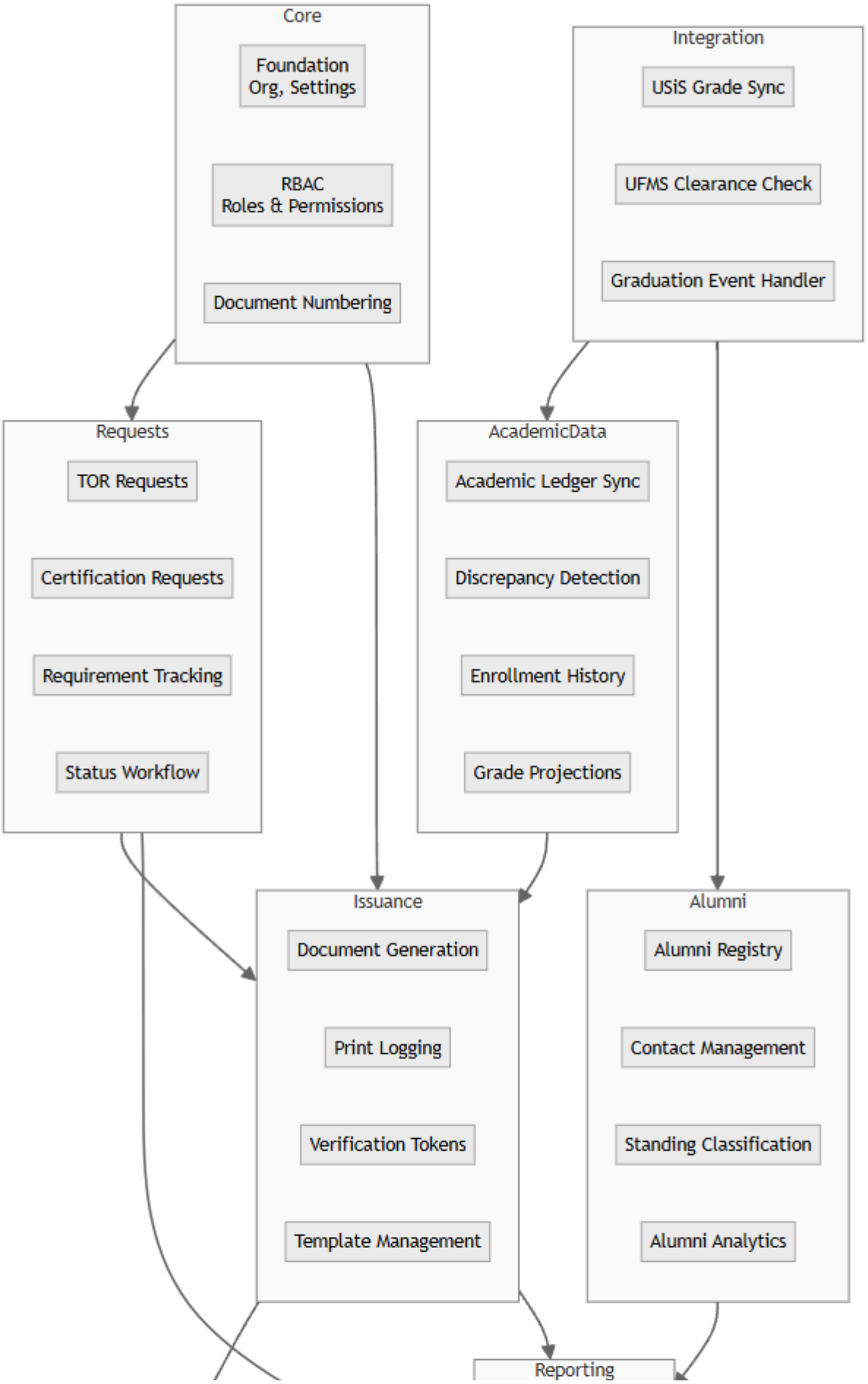
Deployment notes:

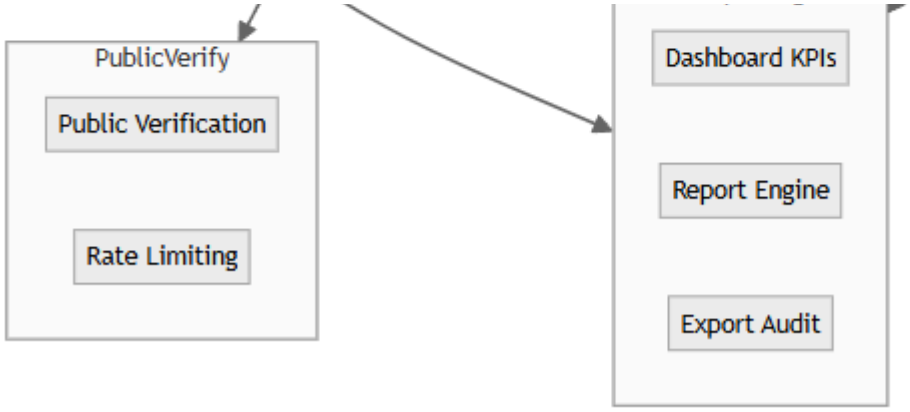
- Records web served statically from CDN or Vite dev server
- Base API runs in containerized Node.js (non-root)
- PostgreSQL schema `records` isolated within `wmsu_cis` database
- Redis persistent (AOF + RDB) for queue durability
- Worker pool for concurrent job processing

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4. Domain Module Architecture

TORMS records module organized by official records lifecycle:


Core
Foundation
Org, Settings
RBAC
Roles & Permissions
Document Numbering
Integration
USiS Grade Sync
UFMS Clearance Check
Graduation Event Handler
Requests
TOR Requests
Certification Requests
Requirement Tracking
Status Workflow
AcademicData
Academic Ledger Sync
Discrepancy Detection
Enrollment History
Grade Projections
Issuance
Document Generation
Print Logging
Verification Tokens
Template Management
Alumni
Alumni Registry
Contact Management
Standing Classification
Alumni Analytics
Reporting



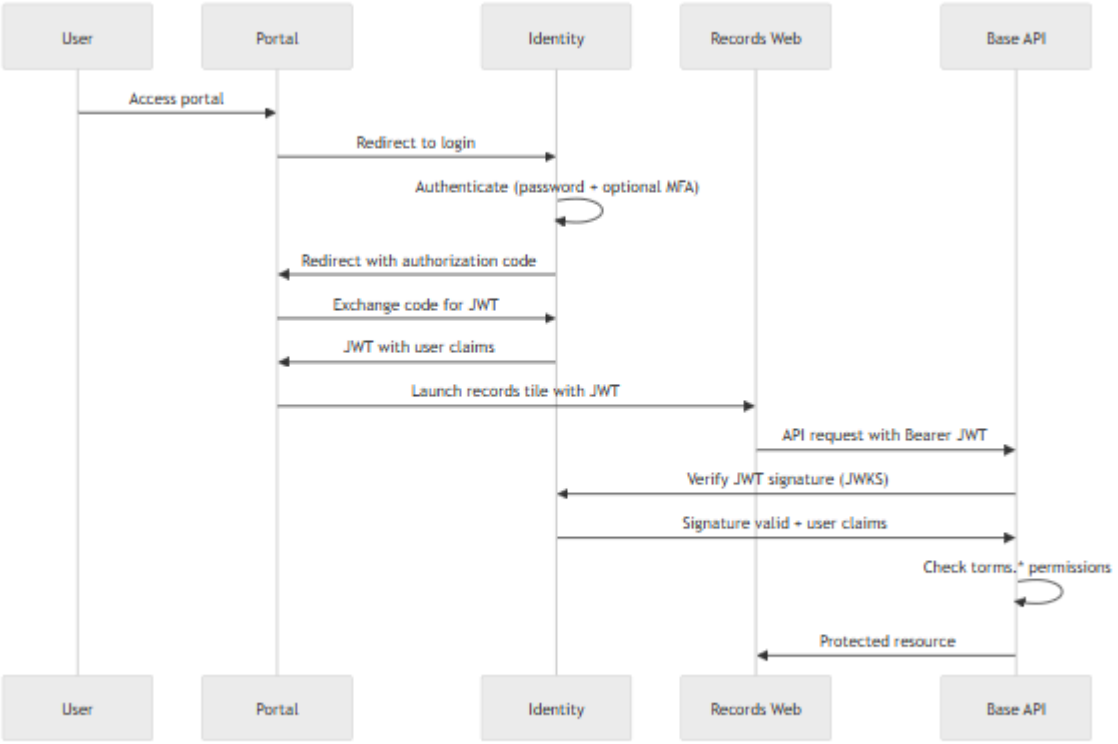
Domain responsibilities:

Domain	Key entities	Services
Core	Organization, Campus, DocumentNumbering, Settings	Foundation config, numbering schemes
Requests	TORRequest, CertificationRequest, RequestRequirement, StatusHistory	Request lifecycle management
Issuance	Issuance, IssuanceLog, VerificationToken, PrintEvent	Document generation, verification
Academic Data	LedgerProjection, GradeProjection, Discrepancy, SyncRun	USiS data sync, discrepancy detection
Alumni	AlumniProfile, AlumniContact, AlumniEmployment, AlumniEducation	Alumni registry, standing management
Integration	USiSGradeSync, UFMSClearanceClient, GraduationEventHandler	External system integration
Public Verify	VerificationAttempt, RateLimitRecord	Public verification, rate limiting
Reporting	Report, ReportExport, DashboardMetric	Analytics, export audit

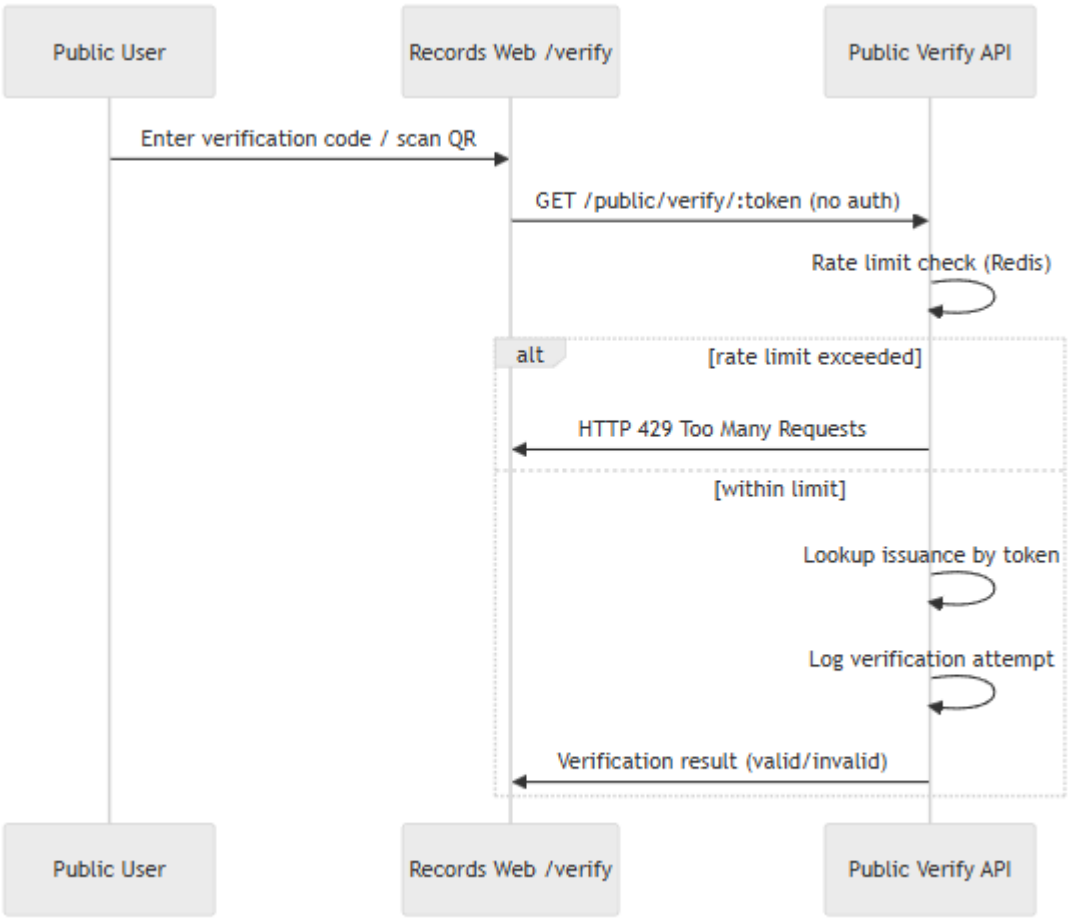
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5. Security Architecture

5.1 Authentication Flow



Public verification flow:



Security controls:

Layer	Control	Implementation
Transport	TLS 1.2+	Reverse proxy termination
Authentication	OIDC/JWT	Identity service RS256 tokens
Authorization	RBAC	`torms.*` permission checks
Public endpoints	Rate limiting	Redis-backed limiter (10/min/IP)
Internal APIs	Service keys	`X-Service-Api-Key` header
Document access	Role-based	Issued PDFs restricted to officers
Verification tokens	UUID v4	Cryptographically unique

5.2 Permission Model

TORMS permission categories:

- `torms.requests.*` — Request management (view, create, transition)
- `torms.issuance.*` — Document generation and printing
- `torms.alumni.*` — Alumni registry management
- `torms.sync.*` — Academic sync operations
- `torms.reports.*` — Report generation and export
- `torms.config.*` — System configuration

Role mappings:

- `records_officer` → All `torms.*` permissions
- `student` → `torms.requests.own`, `torms.issuance.view_own`
- `alumni` → `torms.alumni.own`, `torms.issuance.view_own`
- `registrar` (cross-system) → `torms.sync.view`, `torms.reports.view`

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6. Data Architecture

6.1 Schema Organization

PostgreSQL schema **`records`** contains:

Core tables:

- `organizations`, `campuses`, `settings`
- `document_numbering_schemes`

Request lifecycle:

- `tor_requests`, `certification_requests`
- `request_requirements`, `request_status_history`

Issuance:

- `issuances`, `issuance_logs`, `print_events`
- `verification_tokens`

Academic data (projections from USiS):

- `ledger_projections`, `grade_projections`
- `enrollment_history`
- `discrepancies`, `sync_runs`

Alumni:

- `alumni_profiles`, `alumni_contacts`

- `alumni_employment`, `alumni_education`
- `alumni_standing_history`

Integration:

- `sync_runs`, `clearance_checks`

Verification:

- `verification_attempts` (audit log)

Reporting:

- `report_exports` (audit log)

6.2 Data Sync Strategy

USiS → TORMS sync:

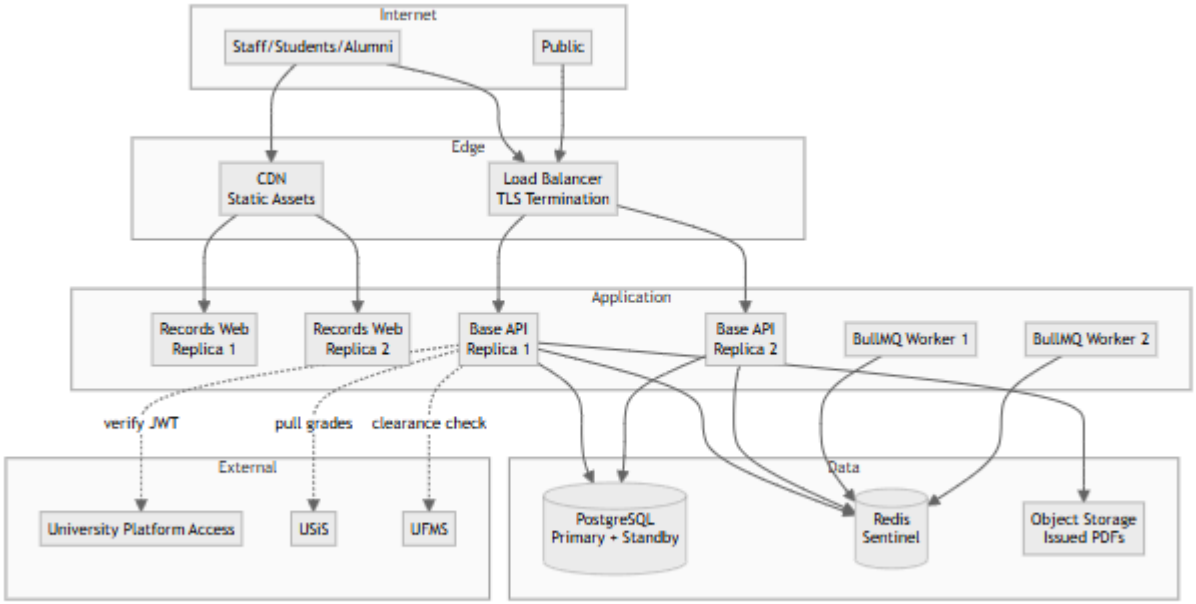
- **Pull model:** TORMS queries USiS internal API
- **Frequency:** Nightly automated job (2:00 AM) + on-demand manual trigger
- **Data:** Grades, enrollment history, graduation events
- **Idempotency:** Upsert by student_id + term + course
- **Discrepancy detection:** Compare projection against USiS source after sync

No reverse sync: TORMS never writes back to USiS. USiS remains single source of truth for academic data.

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7. Deployment Topology

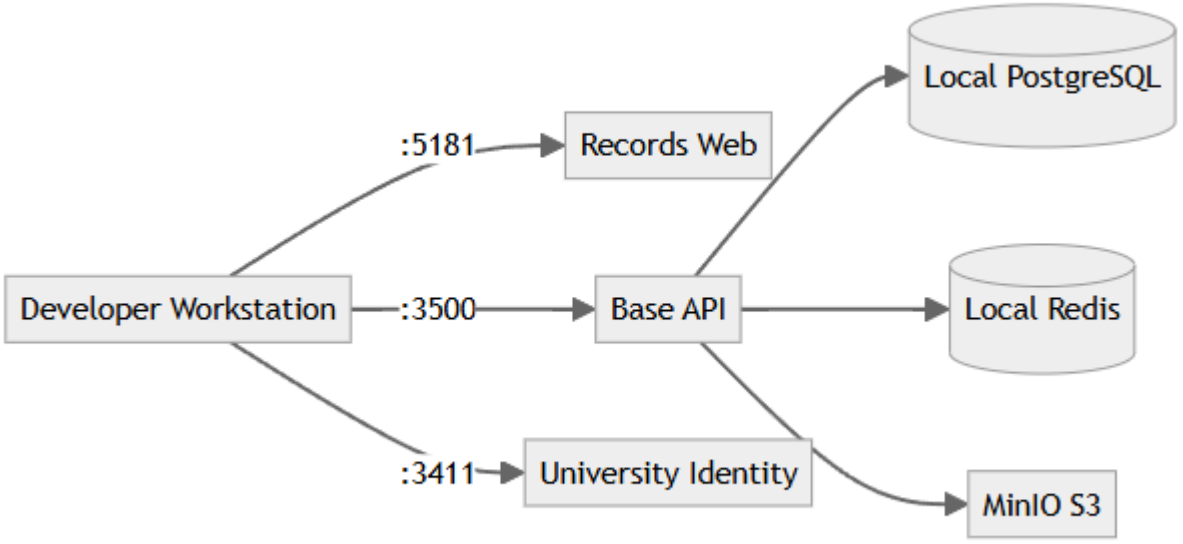
7.1 Production Architecture



Sizing guidance:

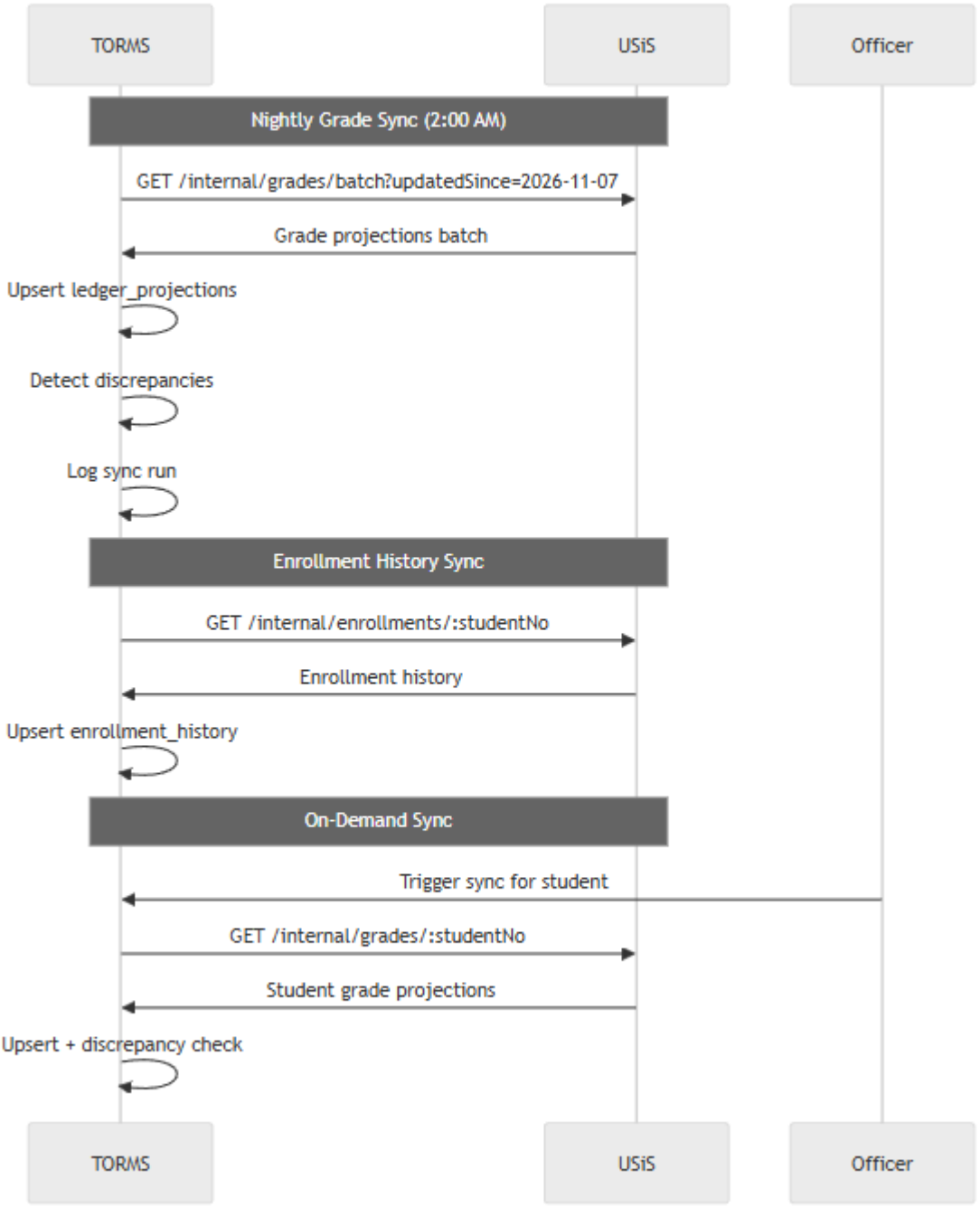
- API: 2 replicas, 2 vCPU / 4 GB RAM each
- Worker: 2 replicas, 1 vCPU / 2 GB RAM each
- PostgreSQL: 4 vCPU / 16 GB RAM
- Redis: 2 vCPU / 4 GB RAM
- Object Storage: S3-compatible managed service

7.2 Local Development

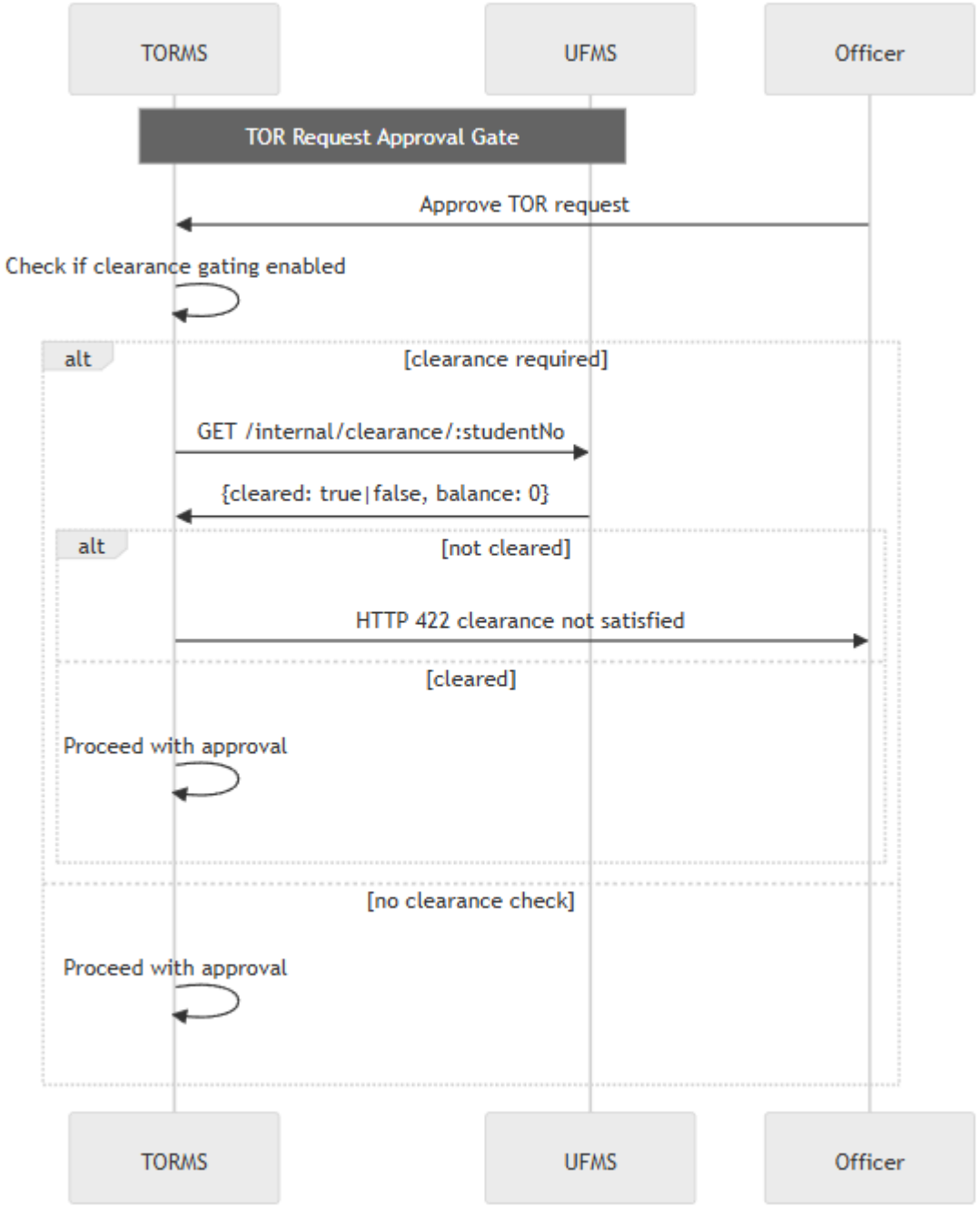


8. Integration Architecture

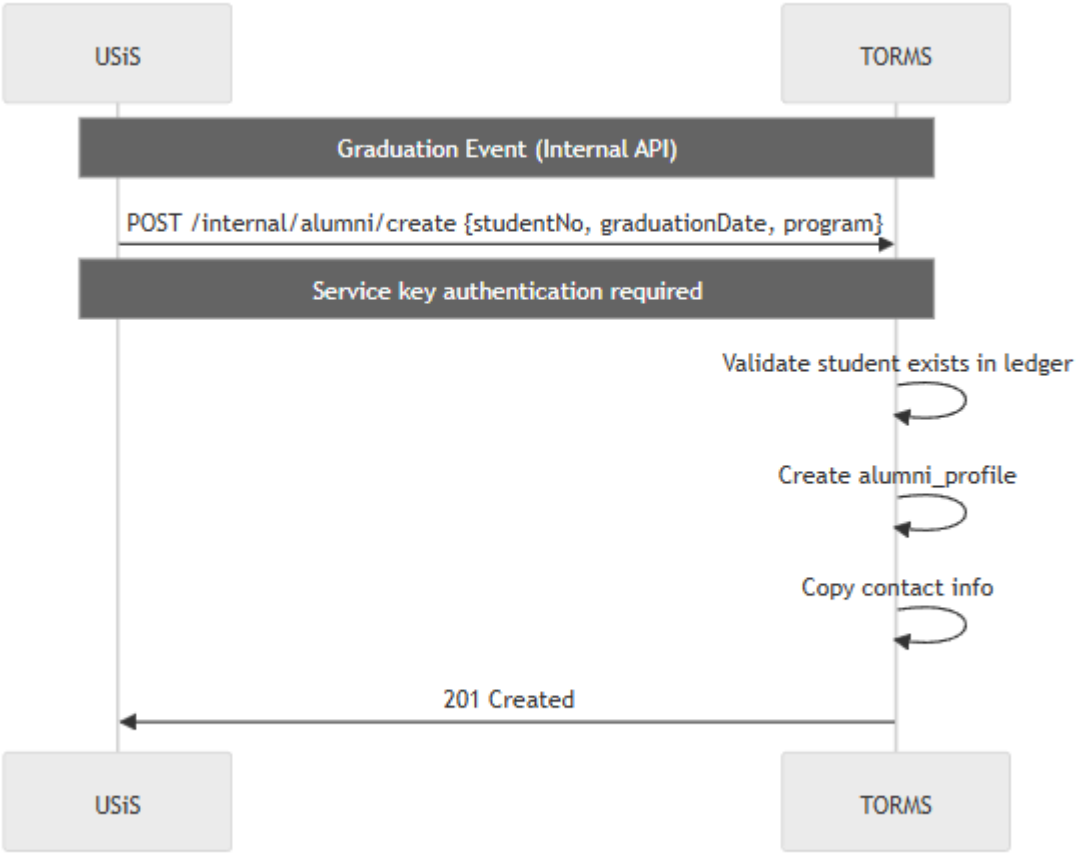
8.1 USiS Academic Data Sync



8.2 UFMS Clearance Check



8.3 Alumni Creation Trigger



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9. Quality Attributes

9.1 Performance

Optimization strategies:

- Database indexing on student_no, status, issue_date, verification_token
- Redis caching for frequently accessed settings (document numbering, templates)
- Background PDF generation (async job queue) for large TOR documents
- Lazy loading in records web UI
- Paginated API responses (default 25, max 100)
- Read replicas for reporting queries

Monitoring:

- APM (Sentry/OpenTelemetry) for request tracing
- PDF generation time metrics
- Sync job duration and error rates
- Public verification endpoint latency

9.2 Scalability

Horizontal scaling:

- Stateless API design
- Load balancer with health checks
- Worker pool for concurrent job processing
- Database connection pooling (PgBouncer)

Data growth:

- Partitioning for large tables (issuances, verification_attempts) by year
- Archival strategy for old verification attempts (retain 1 year)

9.3 Reliability

Failure handling:

- Graceful degradation when USiS unavailable (serve cached ledger projections, warn users)
- Retry logic for sync jobs (exponential backoff)
- Circuit breaker for external API calls
- Immutable issued documents — no deletion, only void with audit trail

Data integrity:

- Foreign key constraints
- Transaction management with rollback
- Idempotent sync operations
- Audit trails for all state changes

9.4 Maintainability

Code organization:

- Domain-driven design with bounded contexts
- Service layer abstraction
- Shared validation schemas
- OpenAPI documentation

Operational:

- Structured JSON logging
- Feature flags
- Blue-green deployment
- Automated schema migrations

9.5 Compliance

Data privacy (RA 10173):

- Student PII access restricted by role
- Public verification limited to name + document metadata
- Audit logging of document access
- Retention policies configurable

Records management:

- Issued documents immutable (void only, never delete)
- 7-year audit trail retention
- Chain of custody preservation for issuances

End of Architecture Design — TORMS

Part 3 — Features & Requirements

TORMS — Transcript & Official Records Management System Features & Requirements

Field	Value
Document ID	FEAT-TORMS-001
Product	TORMS
Companions	TORMS-SRS.md · TORMS-LLD.md
Date	November 2026

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1. Feature Summary

TORMS provides comprehensive official records management for Philippine university compliance:

Area	Features	Status
Authentication	University Platform Access SSO, role-based routing	Delivered
TOR Requests	Submission, status workflow, requirement tracking, PDF generation, print logging	Delivered
Certifications	Multiple types (honorable dismissal, good moral, units, completion), PDF generation	Delivered
Academic Ledger	Grade/enrollment sync from USiS, discrepancy detection, reconciliation	Delivered
Alumni Registry	Profile management, contact updates, standing classification	Delivered
Public Verification	Unauthenticated QR/code verification with rate limiting	Delivered
Dashboard	KPIs, pending requests, recent issuances, sync status	Delivered
Reporting	Issuance reports, request aging, alumni demographics, export audit	Delivered
Configuration	Document numbering, templates, signatories, clearance rules	Delivered

---

2. Feature Catalog

Feature ID	Feature	Description	Priority	Actors

F-AUTH-01	SSO Authentication	University Platform Access OIDC/JWT	Must	All
F-AUTH-02	Role Routing	Home path by role (staff/student/alumni)	Must	All
F-AUTH-03	Permission Enforcement	`torms.*` checks on all endpoints	Must	System
F-TOR-01	TOR Request Submission	Student submits with purpose, delivery mode, copies	Must	Student
F-TOR-02	Request Numbering	Auto-assign unique numbers (TOR-YYYY-NNNN)	Must	System
F-TOR-03	Status Workflow	submitted → requirements_pending → processing → approved → issued → released	Must	Officer, System
F-TOR-04	Request Management	Officer view/filter/search all requests	Must	Officer
F-TOR-05	Status Transitions	Officer advances status with notes	Must	Officer
F-TOR-06	Clearance Validation	Check clearance before approval (configurable)	Must	System
F-TOR-07	Payment Verification	UFMS payment check if gating enabled	Should	System
F-TOR-08	Requirement Attachments	Upload/attach requirement documents	Must	Student, Officer
F-TOR-09	Request Notifications	Email/SMS on status transitions	Should	System
F-TOR-10	TOR PDF Generation	Generate with ledger, seal, signatures	Must	System
F-TOR-11	Verification QR Codes	Embed QR on TOR documents	Must	System
F-TOR-12	Print Logging	Log print events with timestamp/officer	Must	System
F-TOR-13	Digital Download	Student download issued TOR	Must	Student
F-CRT-01	Certification Types	Honorable dismissal, good moral, units, completion, diploma verify	Must	System
F-CRT-02	Certification Requests	Student submit with type and purpose	Must	Student
F-CRT-03	Certification Numbering	Type-specific numbering (HD-YYYY-NNNN)	Must	System

F-CRT-04	Certification Workflow	submitted → processing → issued → released	Must	Officer, System
F-CRT-05	Certification Configuration	Configure types, templates, requirements	Should	Admin
F-CRT-06	Certification PDF	Generate from templates with verification codes	Must	System
F-CRT-07	Certification Print Log	Log print events	Must	System
F-LED-01	Grade Sync	Pull grades from USiS internal API	Must	System
F-LED-02	Ledger Projections	Create/update projections (no USiS mutation)	Must	System
F-LED-03	Discrepancy Detection	Compare projections to USiS source	Must	System
F-LED-04	Discrepancy Reports	Surface discrepancies to officers	Must	Officer
F-LED-05	Manual Resolution	Officer resolves discrepancies with audit	Must	Officer
F-LED-06	Enrollment History Sync	Sync enrollment records from USiS	Must	System
F-LED-07	Sync Metadata Tracking	Log run timestamp, count, errors	Must	System
F-LED-08	Sync Retry Logic	Exponential backoff on failure	Should	System
F-LED-09	On-Demand Sync	Officer trigger sync for specific student	Should	Officer
F-ALM-01	Alumni Auto-Creation	Create from USiS graduation event	Must	System
F-ALM-02	Contact Updates	Alumni update address/phone/email	Must	Alumni
F-ALM-03	Document Viewing	Alumni view issued TOR/certifications	Must	Alumni
F-ALM-04	Standing Classification	Maintain alumni standing (active, distinguished, etc.)	Should	Officer
F-ALM-05	Employment Tracking	Self-reported employment/education	Should	Alumni
F-ALM-06	Alumni Reports	Demographic reports by graduation year/program	Should	Officer
F-ALM-07	Bulk Import	Import legacy alumni data	Could	Admin
F-VRF-01	Public Verify Page	`/verify` without authentication	Must	Public

F-VRF-02	Code/QR Entry	Enter code or scan QR	Must	Public
F-VRF-03	Verification Result	Return valid (name, type, date) or invalid	Must	System
F-VRF-04	PII Protection	No sensitive data beyond name/type/date	Must	System
F-VRF-05	Verification Logging	Log attempts with IP/timestamp	Must	System
F-VRF-06	Rate Limiting	10 requests/minute/IP	Must	System
F-VRF-07	Branded Display	Show university branding on results	Should	System
F-DSH-01	Officer Dashboard	Pending requests, recent issuances, sync status	Must	Officer
F-DSH-02	KPIs	Requests by status, processing time, issuance volume	Should	Officer
F-RPT-01	TOR Issuance Report	By date range, program, student type	Must	Officer
F-RPT-02	Certification Report	By type and date range	Must	Officer
F-RPT-03	Request Aging Report	Days in each status	Must	Officer
F-RPT-04	Alumni Demographics	By graduation year, program, location	Should	Officer
F-RPT-05	Report Export	CSV and PDF formats	Must	Officer
F-RPT-06	Export Audit Log	Log exports with user/timestamp	Must	System
F-CFG-01	Numbering Schemes	Configure prefix, year format, sequence	Must	Admin
F-CFG-02	Template Upload	PDF-compatible document templates	Should	Admin
F-CFG-03	Signatory Config	Authorized signatories with titles/signatures	Must	Admin
F-CFG-04	Clearance Rules	Configure requirements per document type	Should	Admin
F-CFG-05	Config Audit Log	Log configuration changes	Must	System

3. Requirements Traceability Matrix

3.1 Functional Requirements Mapping

Requirement ID	Requirement	Related Features	Priority	Implementation Status
FR-AUTH-001..005	Authentication and authorization	F-AUTH-01..03, F-VRF-01	Must	Delivered
FR-TOR-001..013	TOR request management	F-TOR-01..13	Must/Should	Delivered
FR-CRT-001..007	Certification management	F-CRT-01..07	Must/Should	Delivered
FR-LED-001..009	Academic ledger sync	F-LED-01..09	Must/Should	Delivered
FR-ALM-001..007	Alumni registry	F-ALM-01..07	Must/Should/Could	Delivered
FR-VRF-001..007	Public verification	F-VRF-01..07	Must/Should	Delivered
FR-DSH-001..002	Dashboard	F-DSH-01..02	Must/Should	Delivered
FR-RPT-001..006	Reporting	F-RPT-01..06	Must/Should	Delivered
FR-CFG-001..005	Configuration	F-CFG-01..05	Must/Should	Delivered

3.2 Non-Functional Requirements Mapping

NFR Category	Requirement ID	Related Features	Implementation Approach
Performance	NFR-PERF-001..005	All features	Database indexing, Redis caching, async PDF generation, pagination
Scalability	NFR-SCAL-001..004	All features	Horizontal API scaling, worker pools, table partitioning
Security	NFR-SEC-001..010	F-AUTH-*, F-VRF-*, F-CFG-*	TLS, RBAC, rate limiting, UUID tokens, audit logging, RA 10173 compliance
Reliability	NFR-REL-001..005	All features	HA deployment, backups, DR procedures, graceful degradation, immutable docs
Maintainability	NFR-MAIN-001..005	All features	Modular architecture, OpenAPI, structured logs, versioned migrations

Compliance	NFR-COMP-001..004	Student/alumni data	RA 10173, FERPA-equivalent, 7-year audit retention

3.3 Source Requirements Traceability

WMSU TOR VI.E.6 compliance matrix:

TOR Requirement	Feature IDs	Implementation Notes	Compliance Status
TOR request and processing	F-TOR-01..13	Full lifecycle from submission to issuance	Delivered
Multiple certification types	F-CRT-01..07	Honorable dismissal, good moral, units, completion, diploma verify	Delivered
Academic record management	F-LED-01..09	USiS sync with discrepancy detection; TORMS is projection, not SoR	Delivered
Alumni registry	F-ALM-01..07	Auto-creation from graduation, contact management, standing	Delivered
Document verification	F-VRF-01..07	Public QR/code verification with rate limiting	Delivered
Clearance integration	F-TOR-06, F-TOR-07	UFMS clearance checks when enabled	Delivered
Print audit trails	F-TOR-12, F-CRT-07	Print event logging for accountability	Delivered
Reporting and analytics	F-RPT-01..06, F-DSH-01..02	Issuance, aging, alumni reports with export	Delivered

Deferred/Out of scope:

- Grade entry and enrollment workflows → USiS responsibility
- Financial transactions and cashiering → UFMS responsibility
- Library, documents, inventory → ULMIS, UDMS, UIMS

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4. Non-Functional Requirements Summary

4.1 Performance Targets

Metric	Target	Measurement
API response (p95)	≤2s standard queries	APM monitoring
PDF generation	≤10s per TOR	Performance profiling

Public verification (p95)	≤1s	Load testing
Academic sync (1000 students)	≤30 minutes	Job duration metrics

4.2 Security Controls

Control	Implementation	Verification
TLS enforcement	Reverse proxy/ingress	SSL Labs scan
RBAC	`torms.*` permission checks	Automated security tests
Rate limiting	Redis-backed (10/min/IP)	Load test verification
Audit logging	Structured logs for issuance/print/config	Log review
Token uniqueness	UUID v4	Code review + statistical testing

4.3 Reliability Measures

Measure	Target	Method
Availability	≥99% business hours	SLA monitoring
Backup frequency	Daily incremental, weekly full	Automated verification
RPO	≤15 minutes	DR drill testing
Document immutability	No deletion post-issuance	Business rule enforcement + audit

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5. Actors & Feature Access

Actor	Primary Features	Restricted Features
Records Officer	F-TOR-04..12, F-CRT-05..07, F-LED-04..05, F-LED-09, F-ALM-04, F-RPT-01..06, F-DSH-01..02, F-CFG-01..05	Full records management
Student	F-TOR-01, F-TOR-08, F-TOR-13, F-CRT-02	Own requests only
Alumni	F-ALM-02, F-ALM-03, F-ALM-05	Own profile and documents
Registrar (cross-system)	F-LED-04, F-RPT-01..06	Oversight and reports
Public	F-VRF-01..07	Verification only
System (Background)	F-TOR-02..03, F-TOR-10..12, F-CRT-06..07, F-LED-01..03, F-LED-06..08, F-ALM-01, F-VRF-03..06	Automated operations

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End of Features & Requirements — TORMS

Part 4 — Low-Level Design (LLD) Workflows

TORMS — Transcript & Official Records Management System Low-Level Design

Field	Value
Document ID	LLD-TORMS-001
Product	TORMS
Companion	TORMS-SRS.md
Date	November 2026

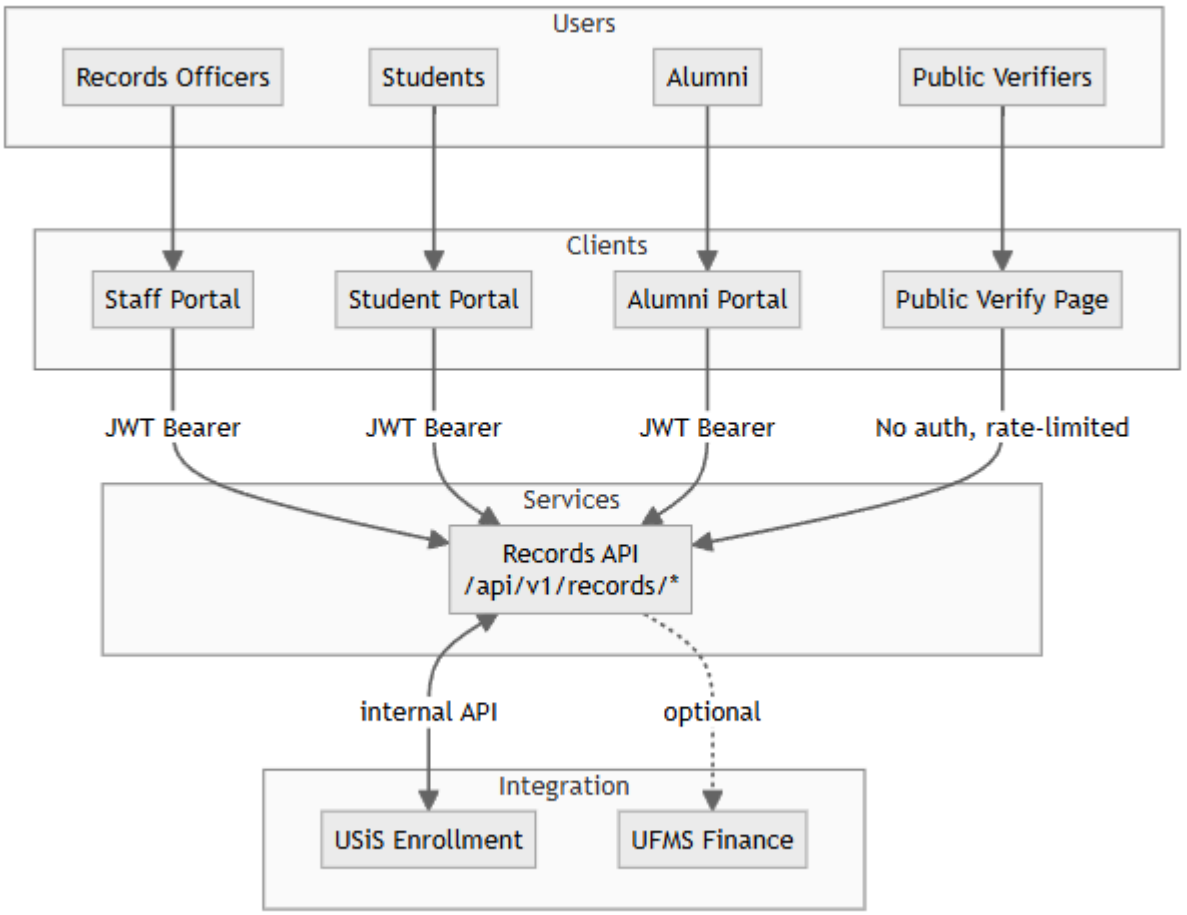
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1. Purpose

This Low-Level Design documents critical workflows for TORMS TOR requests, certification issuance, academic ledger synchronization from USiS, public document verification, alumni management, and records officer operations — providing sequence diagrams, state machines, and implementation details.

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2. System Context (Multi-Role Architecture)



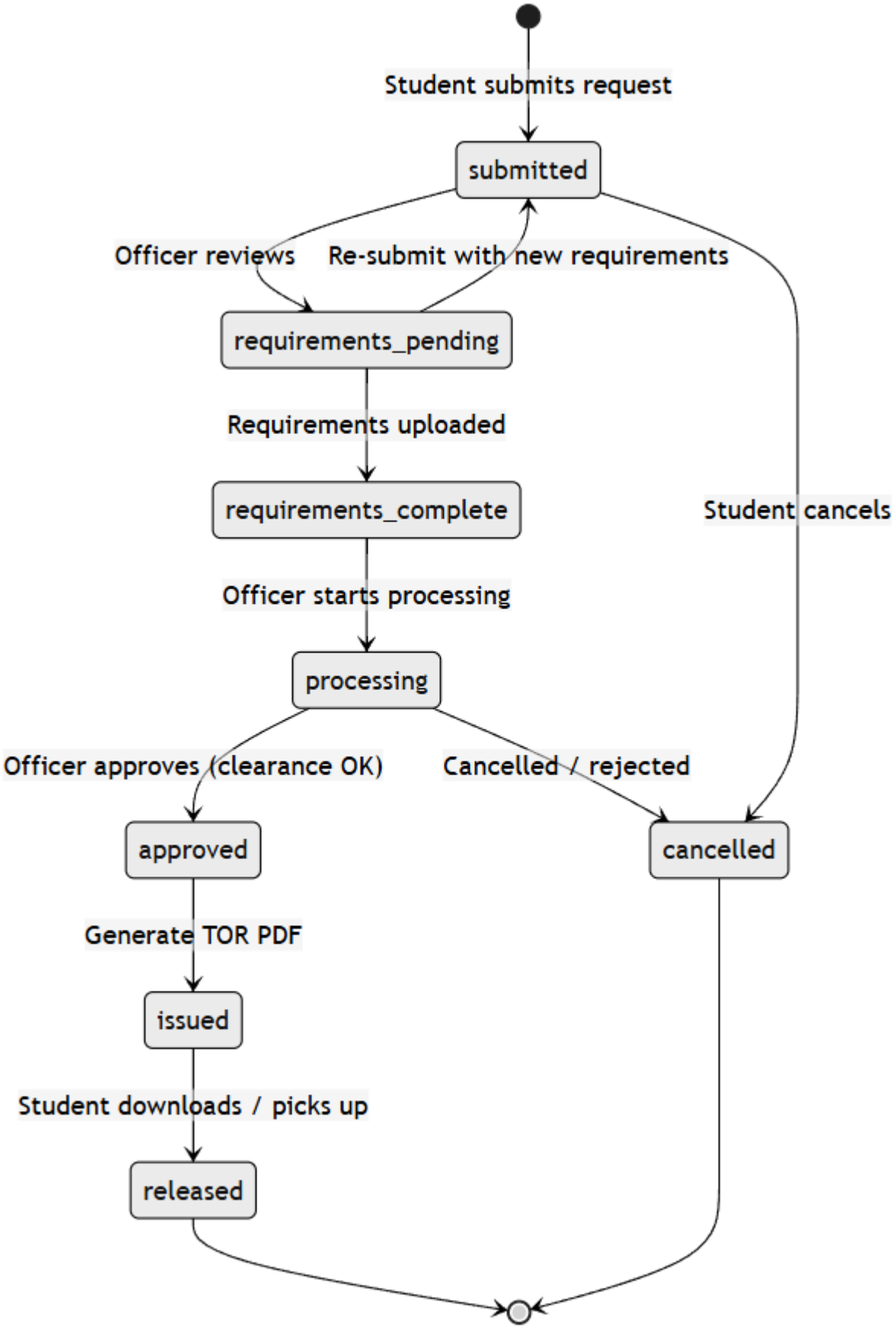
Role-specific routes:

- Records officer: `/staff/*`
- Student: `/student/*`
- Alumni: `/alumni/*`
- Public: `/verify`

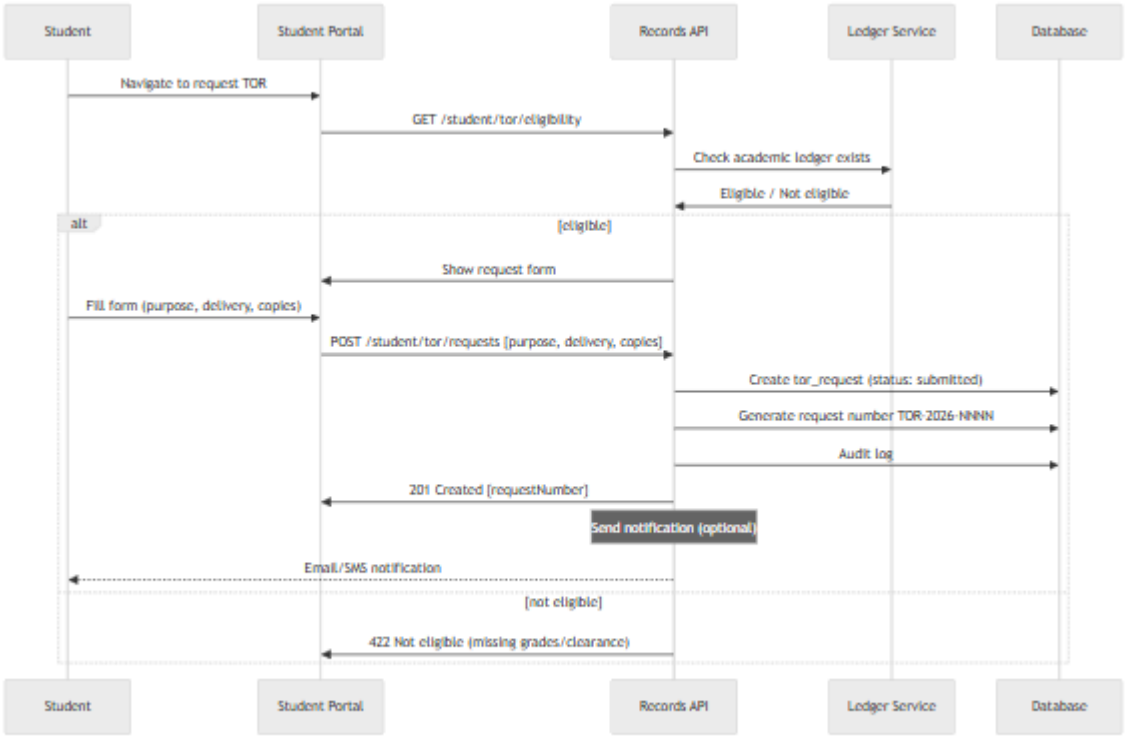
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3. TOR Request Workflow

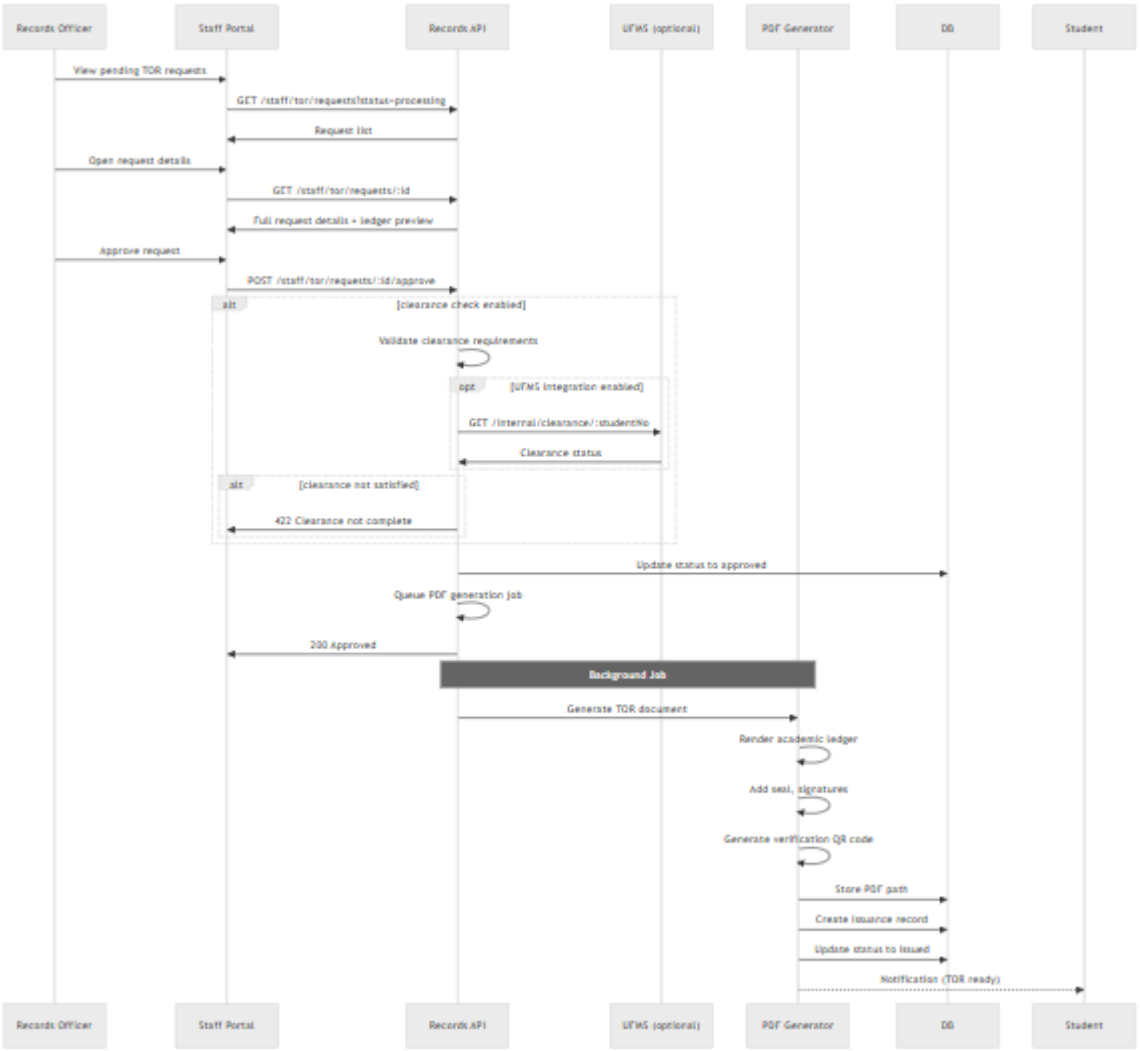
3.1 TOR Request State Machine


Student submits request
submitted
Officer reviews
Re-submit with new requirements
requirements_pending
Requirements uploaded
requirements_complete
Student cancels
Officer starts processing
processing
Officer approves (clearance OK)
Cancelled / rejected
approved
cancelled
Generate TOR PDF
issued
Student downloads / picks up
released

3.2 TOR Request Submission Sequence



3.3 Officer Request Processing

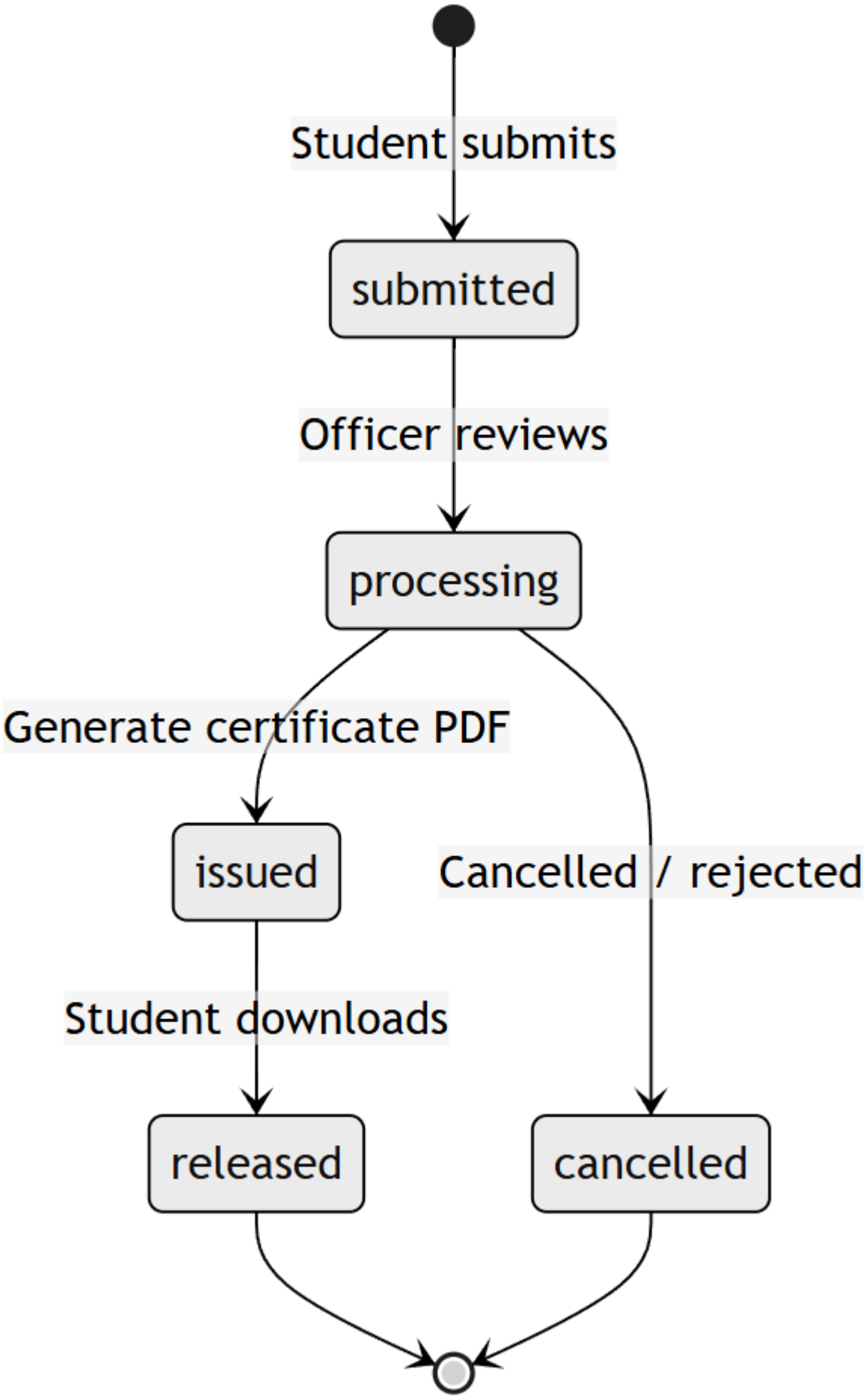


4. Certification Request Workflow

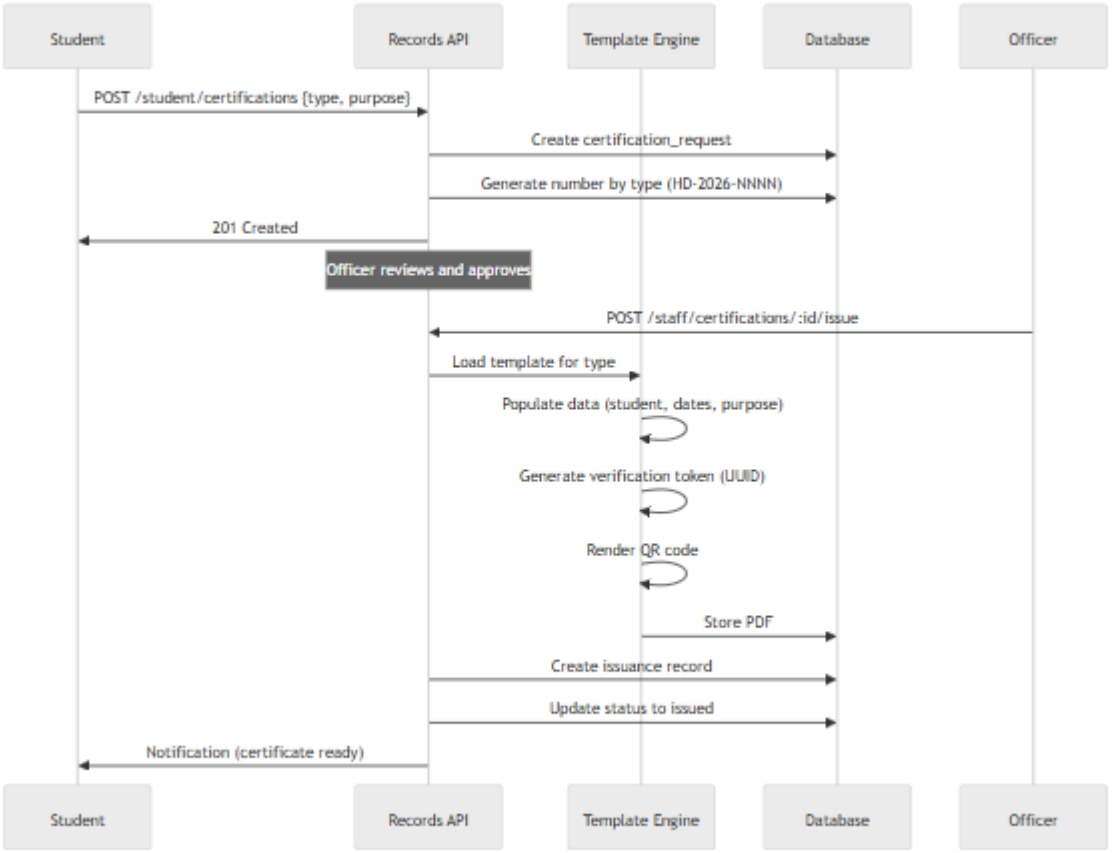
4.1 Certification Types and Numbering

Certification Type	Code	Number Format	Typical Use
Honorable Dismissal	HD	`HD-2026-0001`	Transfer to another institution
Good Moral Character	GMC	`GMC-2026-0001`	Employment, visa applications
Units Earned	UE	`UE-2026-0001`	Proof of partial completion
Completion	CMPL	`CMPL-2026-0001`	Program completion (non-degree)
Diploma Verification	DV	`DV-2026-0001`	Verify diploma authenticity

4.2 Certification Request State Machine


Student submits
submitted
Officer reviews
processing
Generate certificate PDF
issued
Cancelled / rejected
Student downloads
released
cancelled

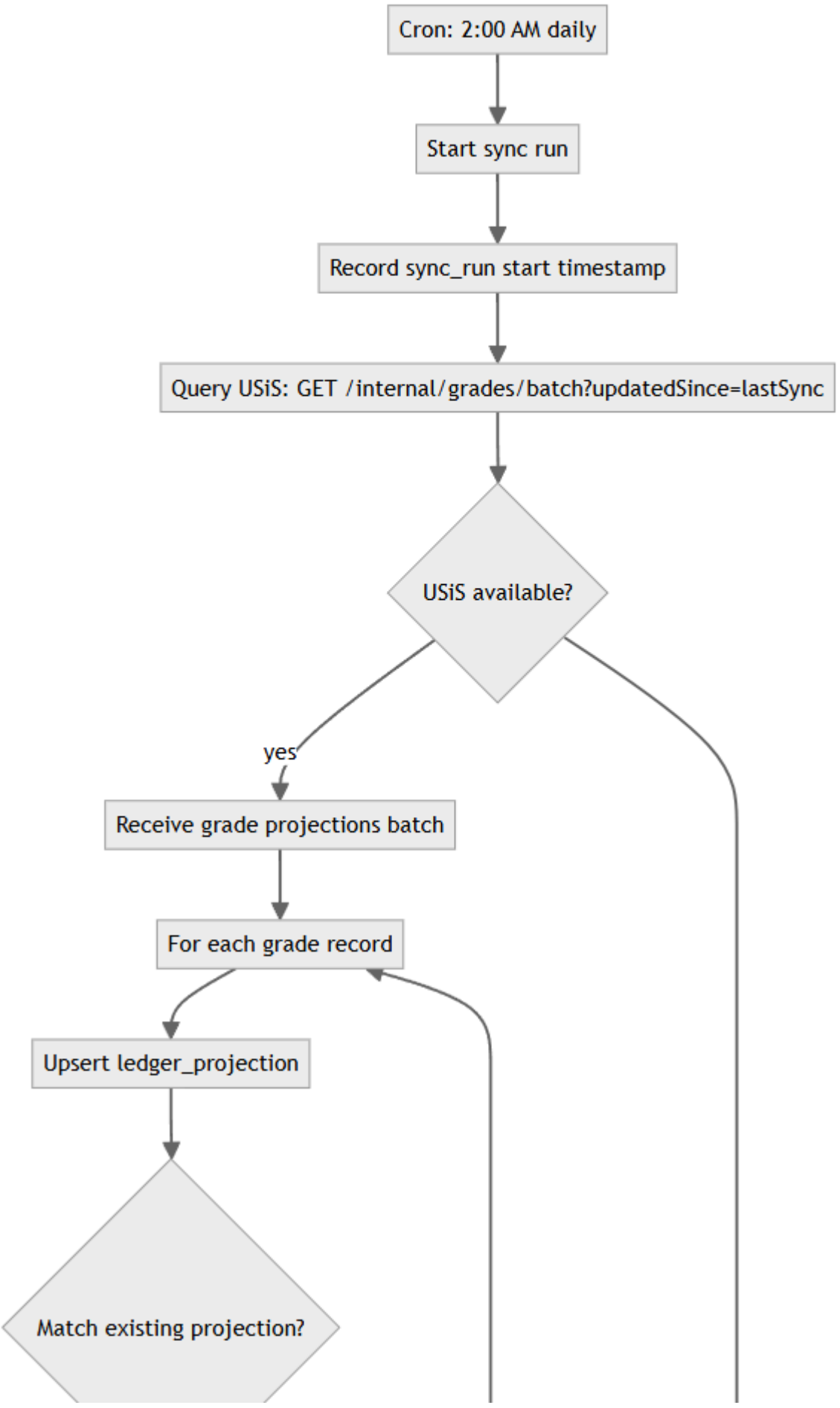
4.3 Certification Issuance Sequence

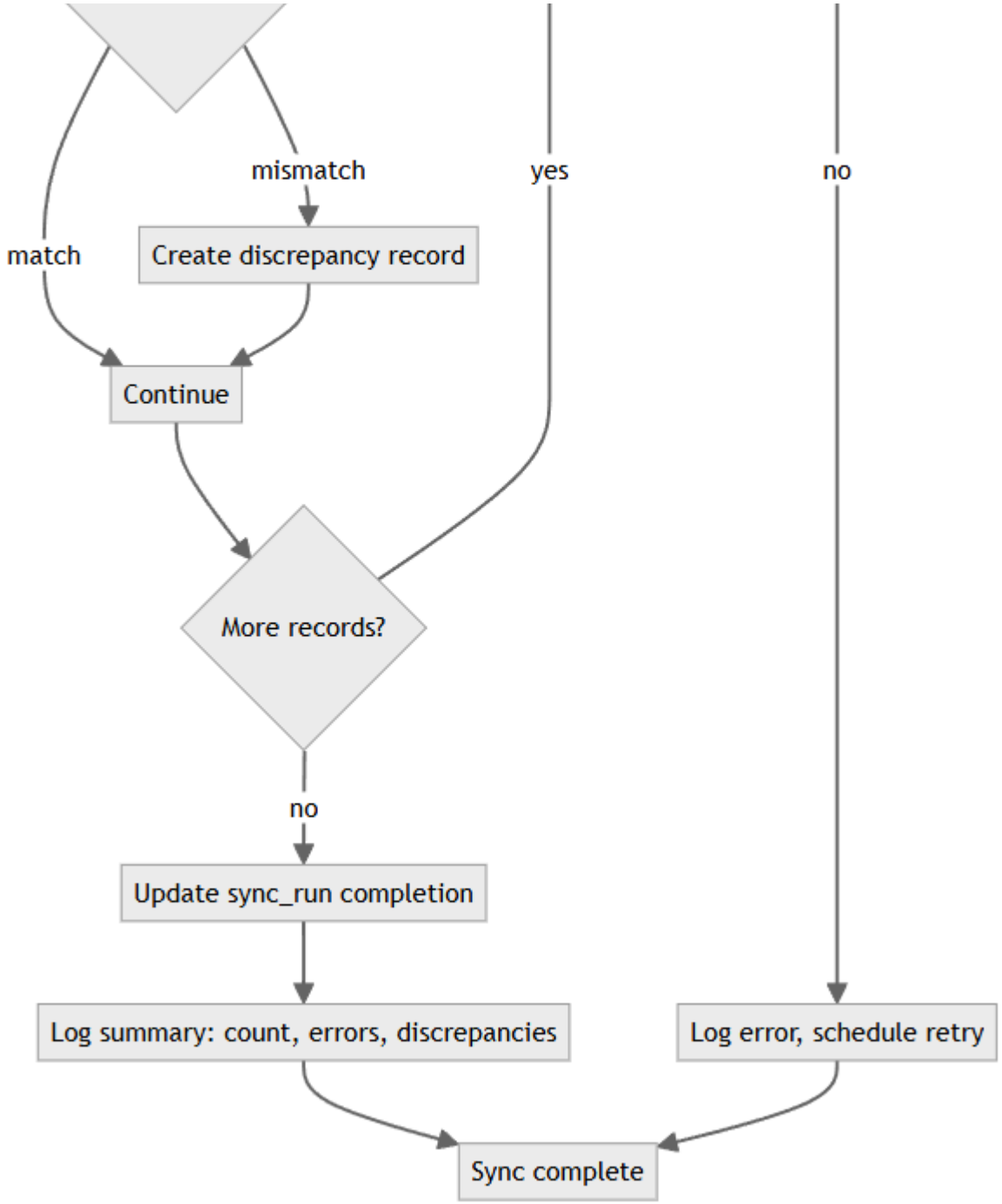


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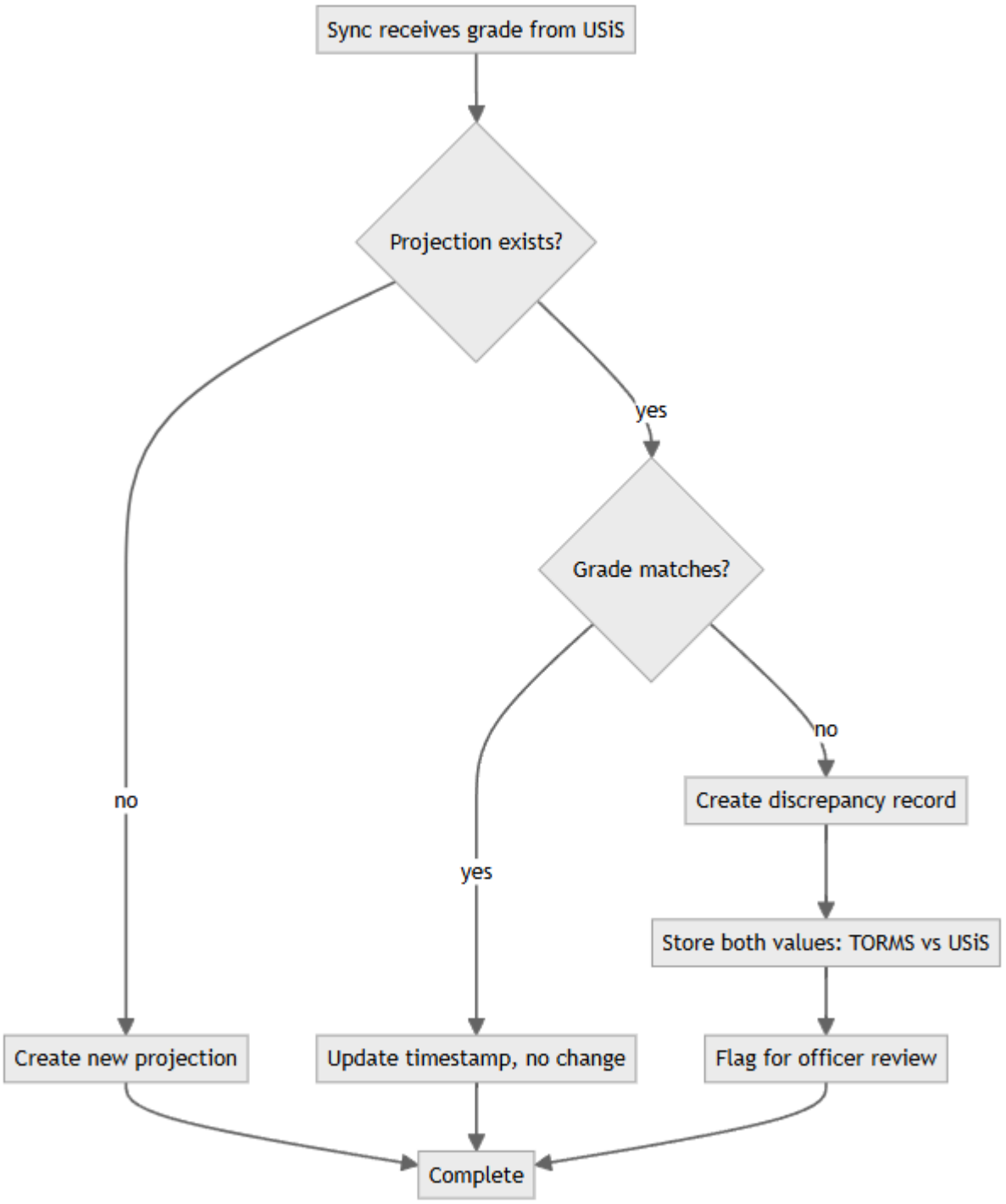
5. Academic Ledger Sync from USiS

5.1 Nightly Batch Sync

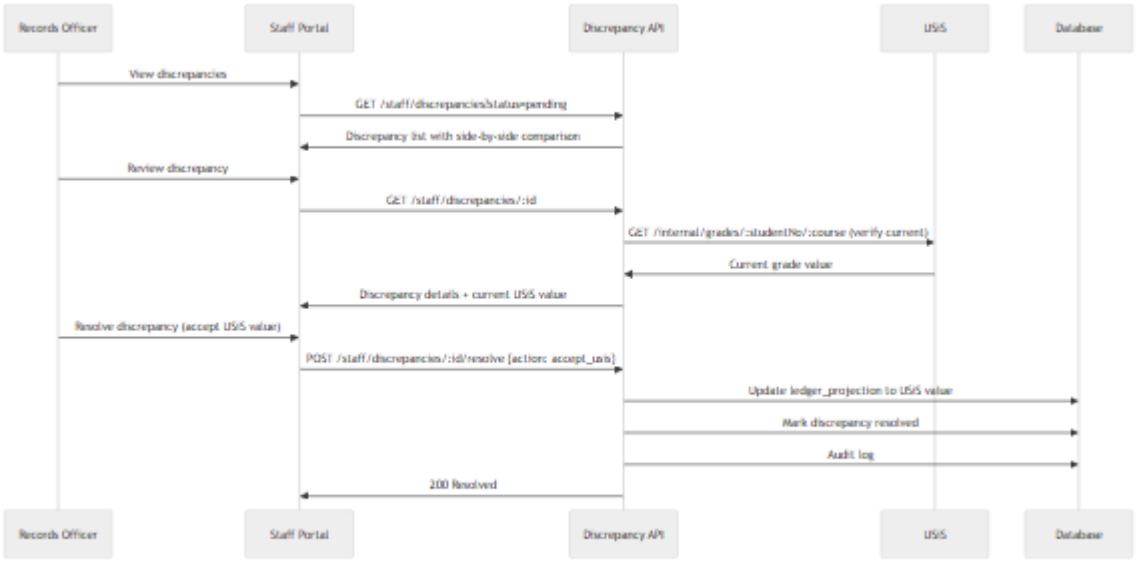

Cron: 2:00 AM daily
Start sync run
Record sync_run start timestamp
Query USiS: GET /internal/grades/batch?updatedSince=lastSync
USiS available?
yes
Receive grade projections batch
For each grade record
Upsert ledger_projection
Match existing projection?



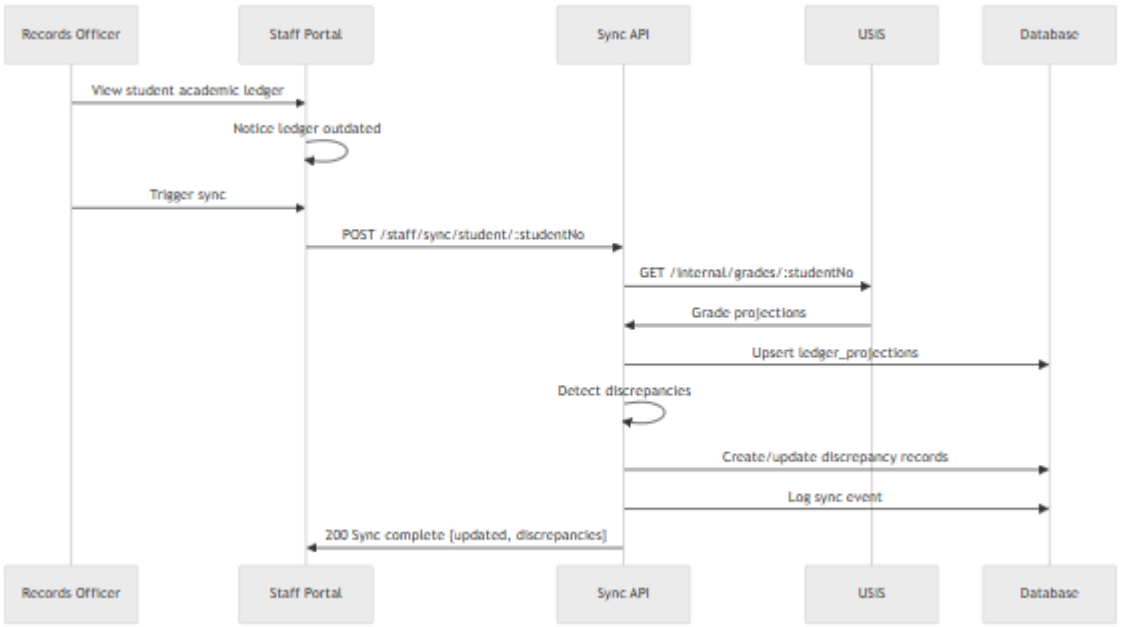
5.2 Discrepancy Detection Logic



5.3 Discrepancy Resolution

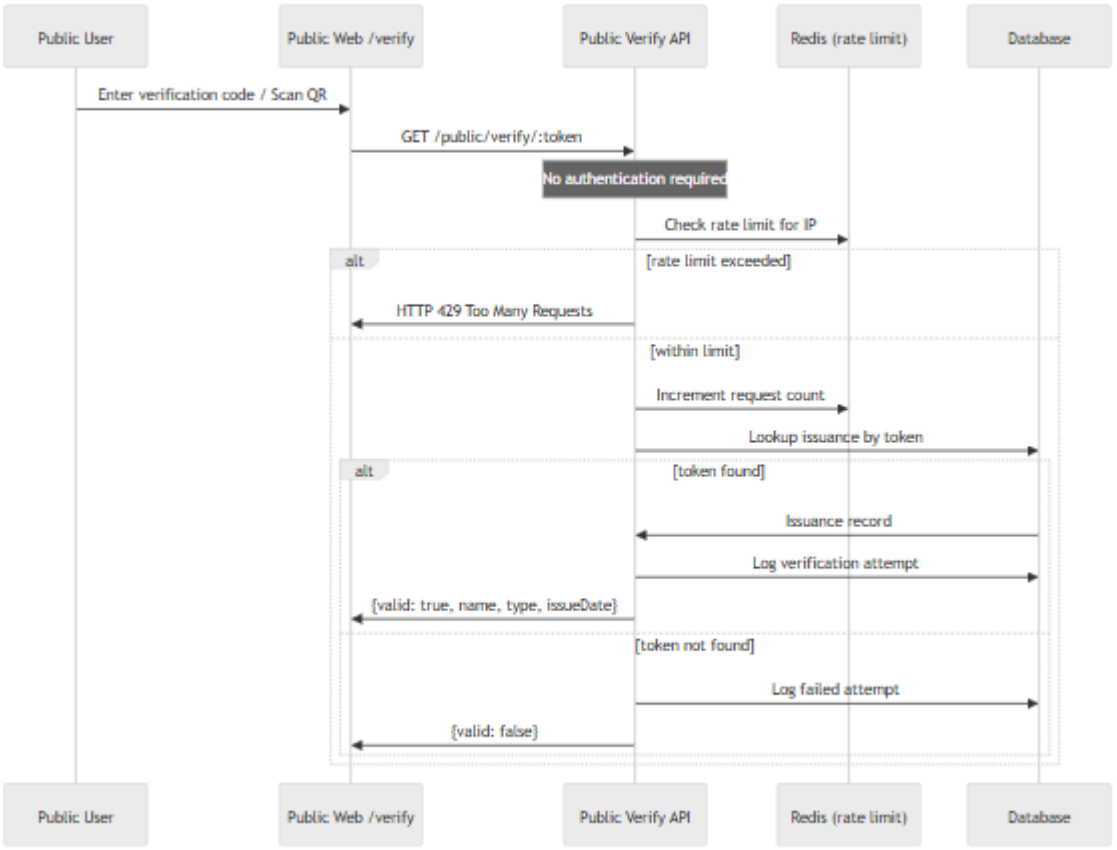


6. On-Demand Sync for Specific Student

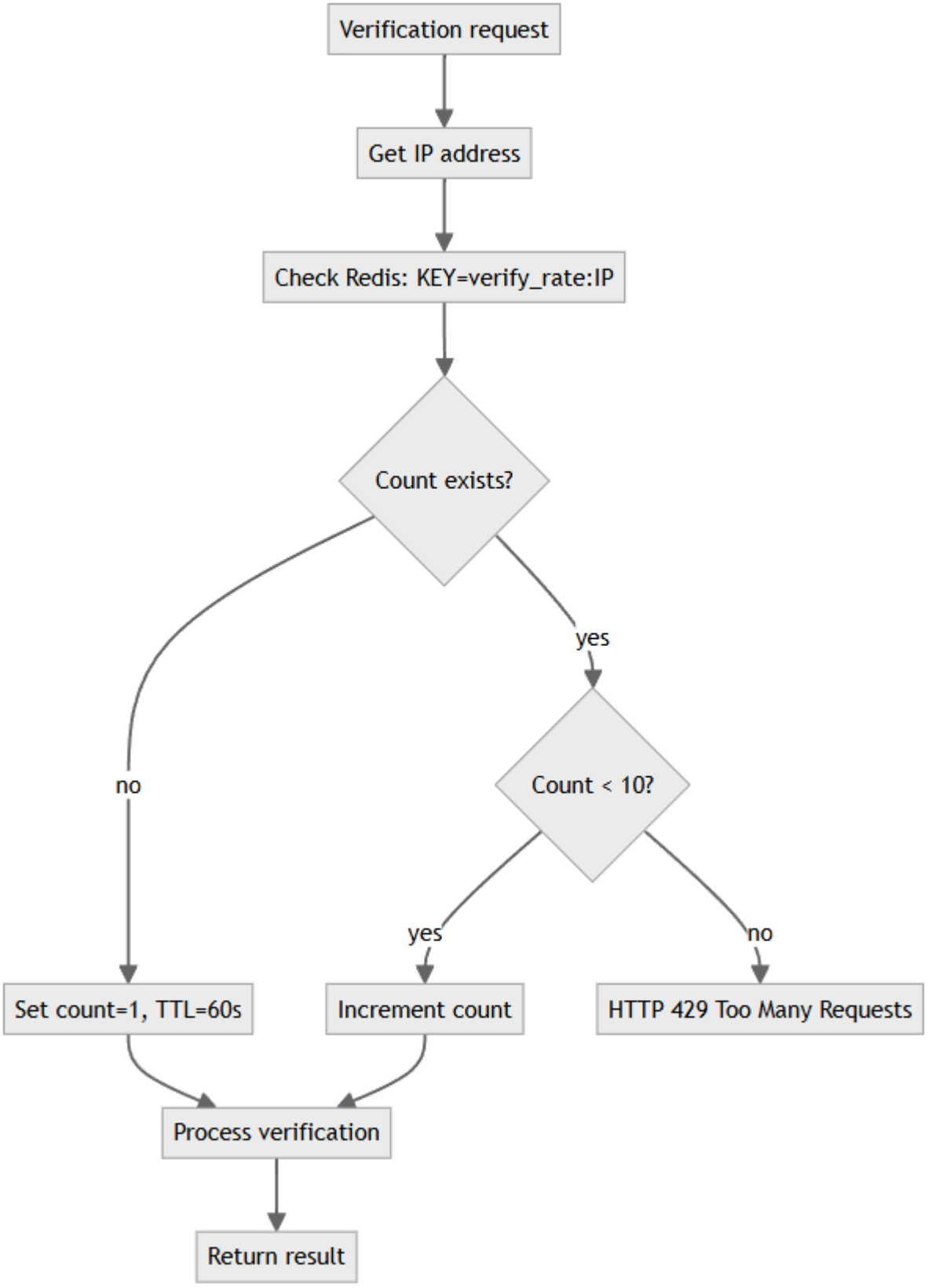


7. Public Document Verification

7.1 Verification Flow

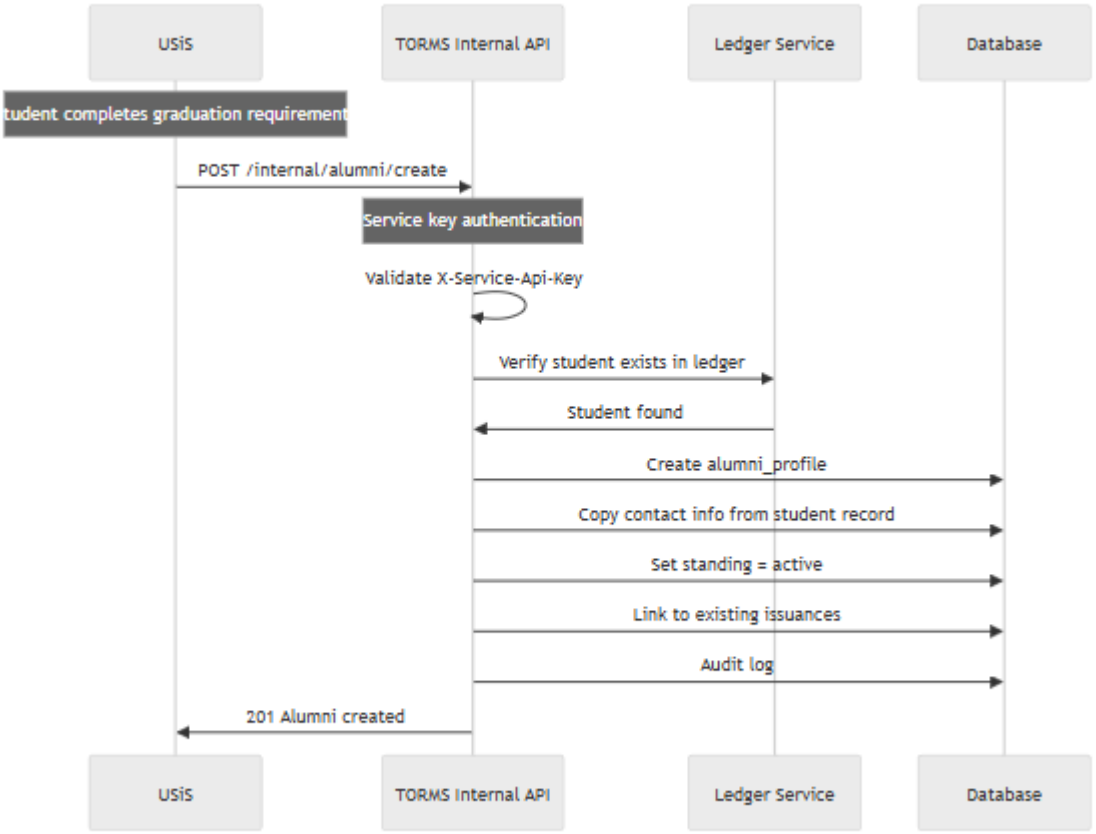


7.2 Rate Limiting Logic

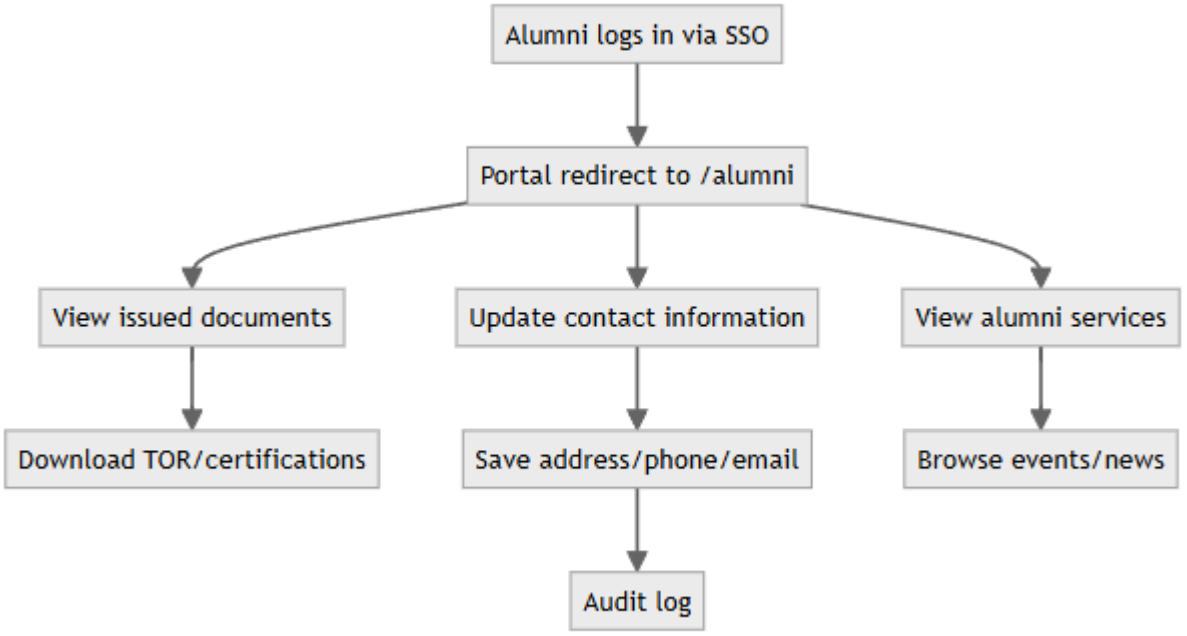

Verification request
Get IP address
Check Redis: KEY=verify_rate:IP
Count exists?
yes
no
Count < 10?
yes
no
Set count=1, TTL=60s
Increment count
HTTP 429 Too Many Requests
Process verification
Return result

8. Alumni Management

8.1 Alumni Creation from Graduation Event



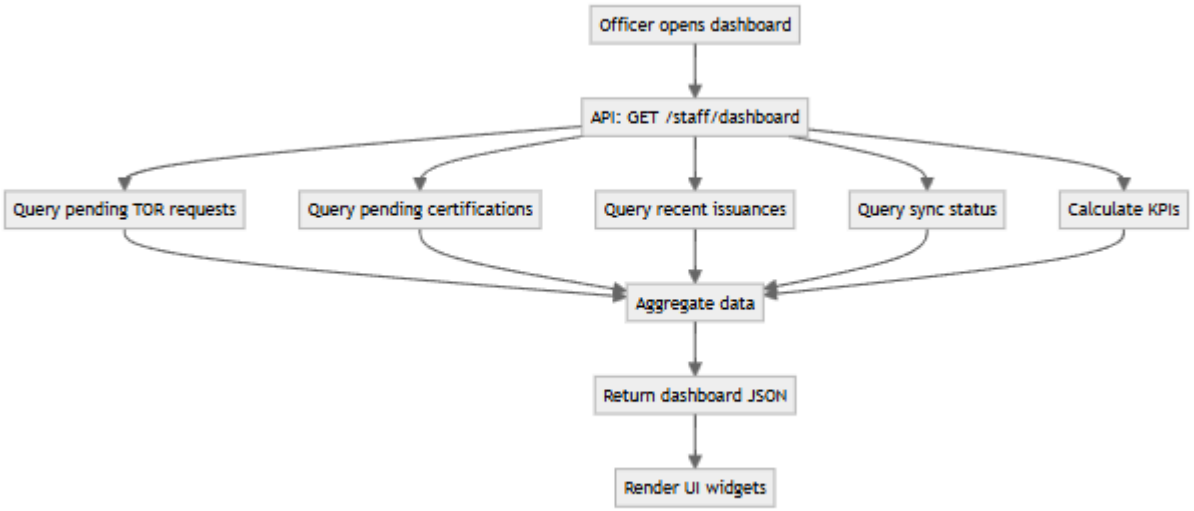
8.2 Alumni Self-Service



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9. Records Officer Dashboard

9.1 Dashboard Data Flow



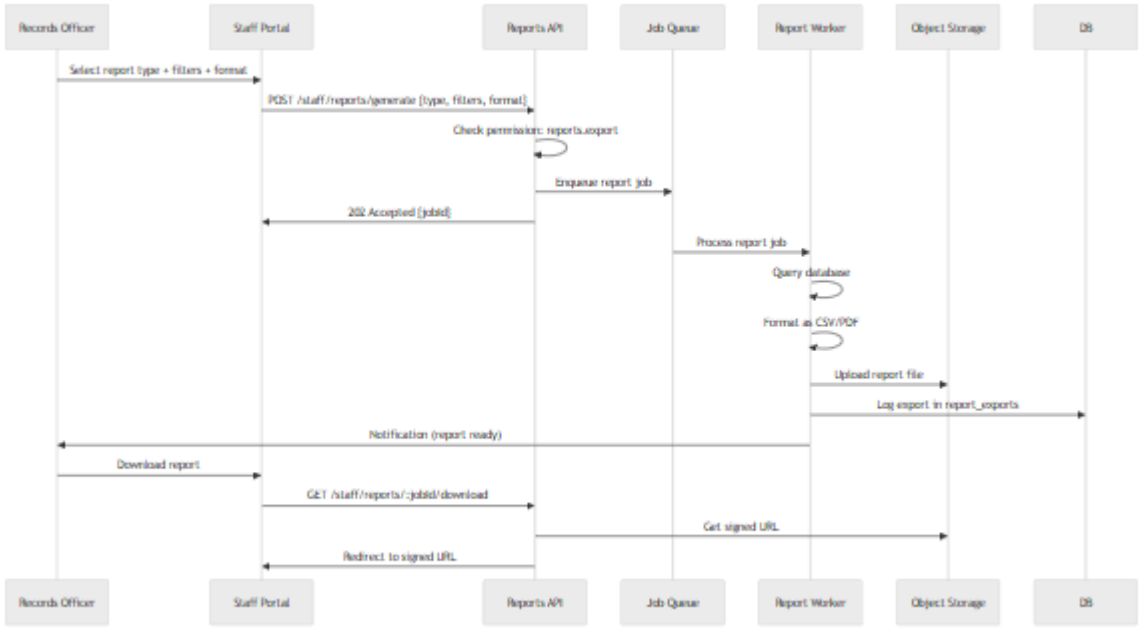
9.2 KPI Calculations

KPI	Calculation	Data source
Total pending requests	Count where status IN (submitted, requirements_pending, processing)	tor_requests + certification_requests
Average processing time	AVG(issued_at - submitted_at) for last 30 days	issuances
Issuance volume (monthly)	Count issuances grouped by month	issuances
Sync status	Latest sync_run timestamp + error count	sync_runs
Discrepancy count	Count where status = pending	discrepancies
Alumni registered (YTD)	Count created this year	alumni_profiles

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10. Report Generation

10.1 Report Export Workflow



10.2 Report Types

Report Type	Query complexity	Typical runtime
TOR Issuance	Simple (issuances + student join)	<5s

Certification Issuance	Simple (issuances + type filter)	<5s
Request Aging	Medium (status history aggregation)	10-30s
Alumni Demographics	Medium (alumni + program join, grouping)	10-30s
Sync Status	Simple (latest sync_runs)	<2s

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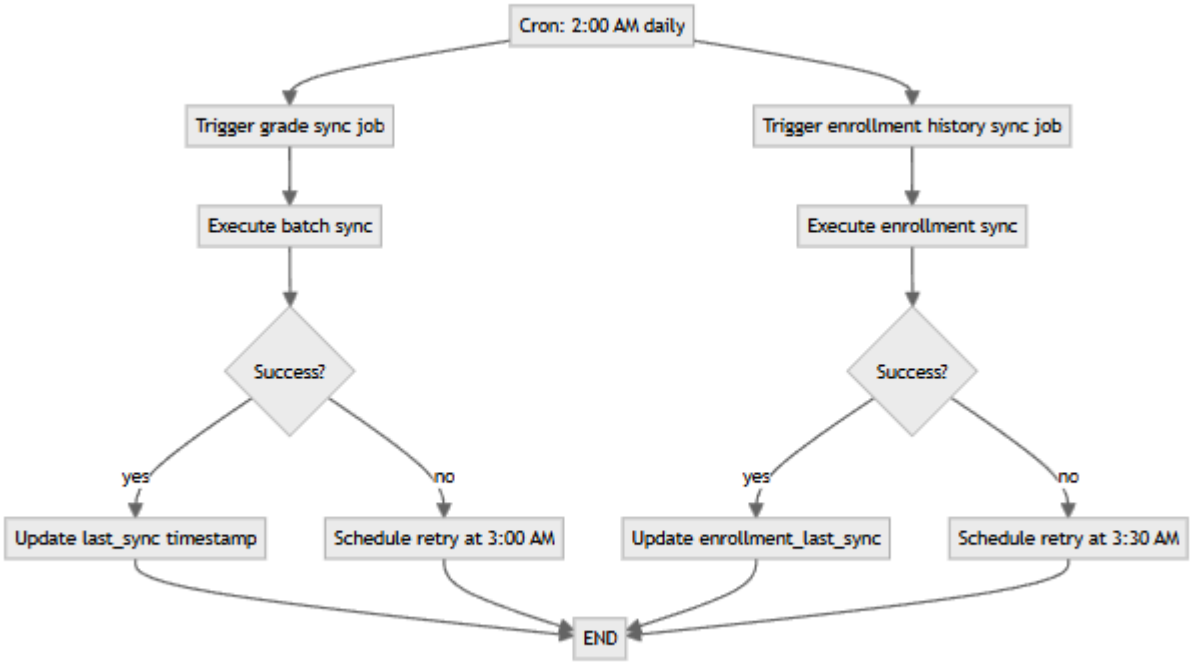
11. Error Cases & Edge Conditions

Case	Expected Behavior	HTTP Status	User Message
TOR request without academic ledger	Block submission	422	"Academic records not found. Contact registrar."
Approve TOR with pending clearance	Block approval	422	"Clearance requirements not satisfied."
UFMS unavailable during clearance check	Log error, allow manual override	503	"Clearance service unavailable. Manual verification required."
Public verify with invalid token	Return not found	404	"Invalid verification code."
Public verify exceeds rate limit	Block request	429	"Too many verification attempts. Try again in 1 minute."
Sync while USiS down	Log error, retry scheduled	N/A	"Sync failed. Will retry automatically."
Duplicate alumni creation (graduation replay)	Idempotent — return existing	200	Existing alumni ID
Student accesses other student's TOR request	Authorization failure	403	"Access denied."
Alumni updates to restricted fields (e.g., graduation date)	Block update	422	"Field cannot be modified."
PDF generation failure	Log error, flag for manual generation	500	"Document generation failed. Request logged."
Discrepancy resolve without permission	Authorization failure	403	"Insufficient permissions."

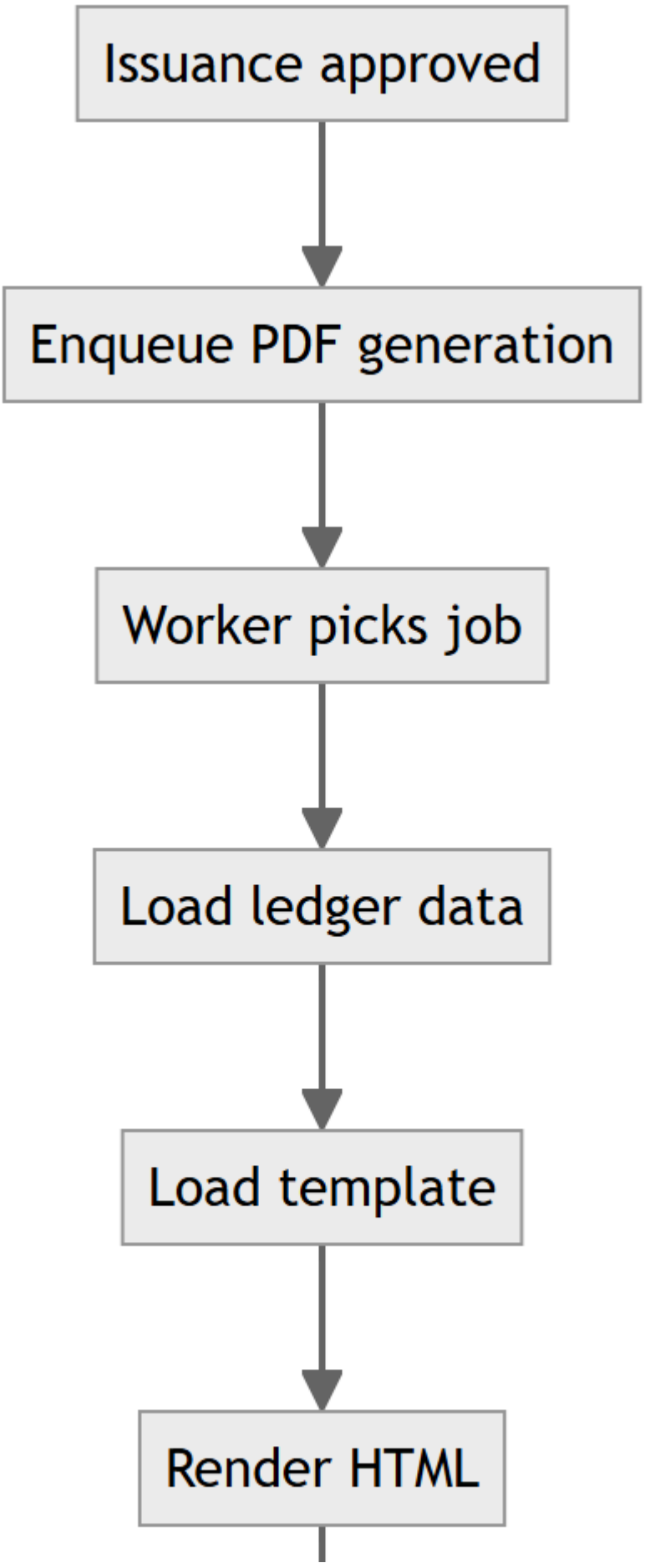
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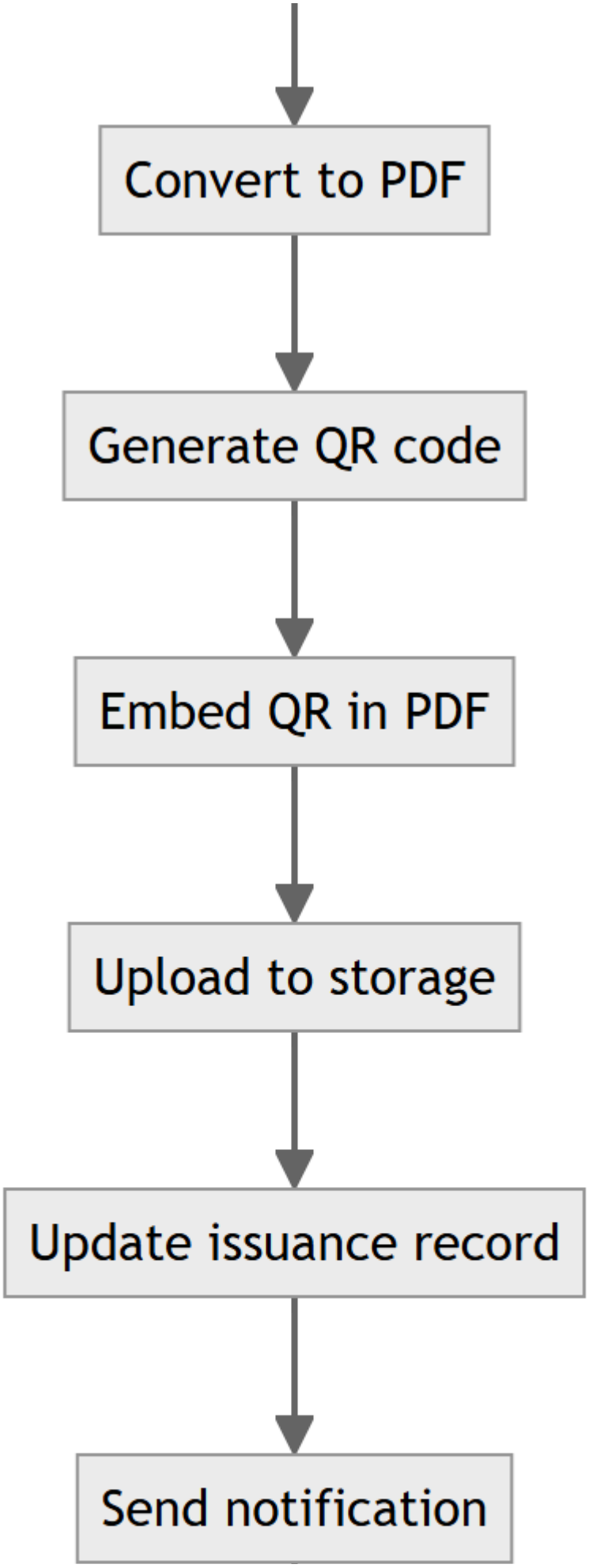
12. Background Job Workflows

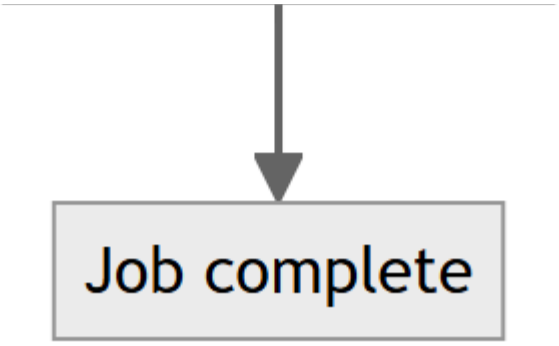
12.1 Nightly Sync Job Schedule



12.2 PDF Generation Job


Issuance approved
Enqueue PDF generation
Worker picks job
Load ledger data
Load template
Render HTML


Convert to PDF
Generate QR code
Embed QR in PDF
Upload to storage
Update issuance record
Send notification



End of LLD — TORMS
